Real-time Target Speaker Extraction

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Abstract

This work builds a causal, real-time, single-channel target speaker extractor whose output is judged by a live speech-to-speech model, Gemini. This work analyses how to build an extractor whose output improves intelligibility for a live model, rather than optimising for signal separation quality. Two errors are considered: words leaked from the interfering speaker and words that neither speaker said that the live model invented or mis-transcribed. An evaluation suite is thus proposed to consider both errors and states the trade-off between them. The extractor built is a causal band-split network, conditioned on an enrollment clip of the target speaker. The extractor is trained on a simulated dataset created from LibriSpeech speakers, with random room acoustics and background noise. Since the live model cannot be differentiated through, a proxy loss is used to train the extractor. Splitting the cue and adding a speaker embedding to the extractor showed improvement over the standard band separation architecture, where the Gemini word error rate dropped from 53.3 % for the baseline to 39.6 % for the extension, while nearly halving the leakage of the interfering speaker's words. The model is seen to perform in real time with a $175\,\mathrm{ms}$ latency on a laptop CPU. The extractor is evaluated on real recordings, holding the same conclusions as the simulated data. Perceptual metrics show how the extractor worsens the perception of the audio relative to the original mixture, but did not indicate the live model's perception of the audio. The extension beats just passing the mixture straight to the live model by dropping the word error rate from 62.94 % to 39.6 %, the rate at which two or more words are leaked in a trial drops from 75.73 % to 33.66 %, the share of the interfering speaker's words reaching Gemini drops from 63.90 % to 23.08 %, but at the cost of increased rate of two or more words fabricated from 30.7 % to 46.9 %. Finally, the extension yields a more stable output than any tested system including an established offline system, WeSep, which is trained on different data and is not optimised for the live model. However, WeSep recovers more of the target when both speakers talk. All results are for at most two-speaker mixtures and are optimised for Gemini.

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Nomenclature

Abbreviations

AMI	Augmented multi-party interaction
ASR	Automatic speech recognition
BAK	Background intrusiveness (DNSMOS P.835 sub-score)
BSRNN	Band-split recurrent neural network
CARTSE	Li and Seki's online REAL-TSE system [1]
DNSMOS	Deep noise suppression mean opinion score
EQ	Equalisation
FR	Fabrication rate
GLU	Gated linear unit
ICR	Interferer content rate
IHM	Individual headset microphone
ITU-T	ITU Telecommunication Standardization Sector
LAM	Large audio model
LCF-WER	Live-model content fidelity word error rate
LSTM	Long short-term memory
LUFS	Loudness units relative to full scale
MLP	Multilayer perceptron
NLTK	Natural Language Toolkit
OVRL	Overall quality (DNSMOS P.835 sub-score)
REAL-TSE	Real-world target speaker extraction (challenge)
RIR	Room impulse response
RTF	Real-time factor
SAR	Signal-to-artefacts ratio
SDR	Signal-to-distortion ratio
SIG	Speech signal quality (DNSMOS P.835 sub-score)
SIR	Signal-to-interference ratio
SI-SDR	Scale-invariant signal-to-distortion ratio
SLT	IEEE Workshop on Spoken Language Technology
SNR	Signal-to-noise ratio
STFT	Short-time Fourier transform
TF Map	Time-frequency map
TSE	Target speaker extraction
VAD	Voice activity detection
WER	Word error rate

Symbols

Signals and transcriptions

x Mixture signal
s Target reference signal
$\hat{s}$ Model output, the estimated target
d Interfering speech signal

Model and objective

X, X_t	STFT of the mixture; frame t of it
X_e, x_e	STFT of the enrollment; the enrollment waveform
S, $\hat{S}$	STFT of the target; of the model output
$\Re(\cdot)$, $\Im(\cdot)$	Real and imaginary parts of a complex value
N, H, $h[n]$	STFT window length, hop and window function
$\tilde{x}_t$, $\tilde{e}_j$	Unit-length mixture frame; unit-length enrollment frame
$C_{t,j}$	Cosine similarity of mixture frame t and enrollment frame j
$A_{t,j}$	Softmax weight of enrollment frame j for mixture frame t
μ_t, u_t	Blended enrollment template; its unit-length direction
m_t, ϕ_t	Projection of $\lvert X_t\rvert$ onto u_t; the baseline cue $m_t u_t$
c_t, r_t	Similarity of $\lvert X_t\rvert$ to u_t; residual $\lvert X_t\rvert - m_t u_t$
M, R	Complex mask; residual spectrogram
α, s_{proj}	Gain of the output along the target; αs
e_{int}, e_{art}	Interference (other speaker and noise); artefact
L_{pres}, L_{MR}, L_{gain}, L_{abs}	Loss terms
w, w_{m}, w_{g}	Loss weights
τ_{pres}, τ_{abs}	Floors of L_{pres} (10^{-3}) and L_{abs} (10^{-2})

AI usage citations

To save space for explanations of how I used AI tools in specific parts of the project, I have made the specific declarations in the section which I reference using tags such as [AI1] and [AI6b].

1. Introduction

Voice assistants engage with users through direct, live speech-to-speech models such as Gemini Live [2]. With the appearance of more voice based applications, the ability to engage with these tools in acoustically challenging environments is becoming increasingly important. A particular problem is the presence of interfering speech, which is misinterpreted by the live speech models as the intended target user speech. Interfering speech thus will confuse the downstream model as to the intended user input, which will lead to a poor user experience. Secondly, live speech models are exposed to noisy environments that include reverberation, background audio and other acoustic challenges. Although live models are fairly robust to background noise on its own (Section 2.4), processing audio to remove interfering speech can introduce artefacts that the live model is not robust to. The work in this report aims to address these issues through the development of a target speaker extractor that aims to preprocess the audio input that is given to the live model. The extractor is constructed under the following constraints: it must be causal, process audio in real time and operate on a single channel of audio. Since the extractor can introduce artefacts that the live model hears as words, it is not enough to measure whether the interfering speaker was removed and thus the evaluation must show which of the two errors the extractor causes.

Under these constraints, the contribution of this work lies not in a new architecture but in how the extractor is evaluated. The novel contributions are:

1. **A live-model content fidelity evaluation suite.** The live model transcribes the extractor's output, and its errors are split into words leaked from the interfering speaker and words that neither speaker said (Section 3.4).

2. **Evidence that the metrics are needed.** A single word error rate hides changes that the split reveals, and an offline transcriber misjudges changes that the live model does not notice (Section 5).

To produce and test these, the following was built:

3. **A constructed dataset and evaluation protocol.** 9955 training trials from 1172 LibriSpeech speakers, each rendered by constructing a simulated room with a random reverberation time and microphone and speaker positions, and then adding real background noise. Each speaker has exact transcripts. The dataset includes trials where the target is absent so that the extractor learns to stay silent (Section 3.1).

4. **A streaming target speaker extractor.** The REAL-TSE baseline architecture [3], implemented for this work and made fully causal, with the enrollment cue split into parts and a speaker embedding added (Section 3.2, Section 3.3). It runs in real time on a laptop CPU within the latency budget and reduces Gemini's word error rate from 53.3 % for the baseline to 39.6 %, while cutting the share of the interfering speaker's words that reach

Gemini almost in half. On 42 validation clips in which only the interfering speaker talks, it falls silent on 18 against 7 for WeSep, a larger offline system trained on different data, although WeSep recovers more of the target when both speakers talk (Section 5.2.1).

5. **A training loop with a proxy objective.** The live model cannot be differentiated through, thus the extractor is trained on a signal-level proxy objective adapted from CARTSE [1].

All results are for two-speaker mixtures and are optimised for Gemini.

The remainder of this report is structured as follows: Section 2 reviews the background information and context for the target speaker extraction domain. Section 3 describes the process of generating the data, the architecture of the extractor and the metrics. Section 4 sets out how the model is trained, selected and evaluated. Section 5 presents the results of the evaluation and Section 6 concludes the report with a summary and potential future work.

2. Literature Review

The section presents and reviews target speaker extraction, streaming separation, how extraction is typically measured, and speech models as listeners [AI1, AI2].

2.1. Target speaker extraction

Target speaker extraction (TSE) is a well established task in speech processing. Meeting transcription systems, hearing aids, and smart speakers, these devices benefit from the ability for more robust speech recognition in the presence of background noise and competing speakers. The main idea is best described by Cherry's cocktail party problem [4]. The problem is a description of a human's ability to focus on a single speaker in noisy environments, and using various indicators such as visual, spatial, and acoustic cues, can understand the speech of a target speaker while ignoring other speakers. The survey done by Žmolíková et al. [5] provides a comprehensive overview, differentiating between three main tasks: target speaker extraction, blind source separation, and noise suppression. In this work, the focus is on target speaker extraction using only voice samples as an acoustic cue.

The intention behind the TSE task is to be able to deploy a system into a real-world scenario. However, due to the variability in the environments that a TSE system may be deployed in, a system must be robust to all conditions and listening scenarios. The survey by Žmolíková et al. [5] highlights the gap that many TSE systems are not trained to go silent when the target speaker is not present, which is a separate task to extraction. The confound of the target speaker always present during training leaves the model vulnerable to situations where there is only interfering speech or noise. Another issue raised by Žmolíková et al. [5] is that the use of signal level metrics such as signal-to-distortion ratio (SDR) is indicative of the ability of the model to extract speech, but *(i)* it does not correlate with audible quality from the perspective of a human listener, and *(ii)* it does not reflect the ability of the model to select or identify a target

speaker as an evaluation metric. The survey also notes that TSE systems are being moved to low latency streaming setups. Therefore it is important to consider lightweight architectures that can be deployed in real-time scenarios.

2.2. Streaming separation

Separation models work with either a raw waveform or spectrogram. Some architectures such as Conv-TasNet [6] favoured the waveform with the reasoning that calculating a spectrogram contributes to the latency of the system. However, Yu et al. adapted a Band-Split RNN (BSRNN) architecture, originally designed for music as an offline model [7], to speech enhancement tasks with an online configuration [8]. Yu et al. found that their online implementation ran faster than real-time on an Intel i5 CPU.

TSE systems must work frame by frame, or chunk by chunk. The REAL-TSE challenge set a benchmark to say that a system must run with an algorithmic latency of $100\,\mathrm{ms}$ or less [3]. Algorithmic latency, separate from end-to-end latency, is the delay built into the model: how long it must wait for audio before it can produce an output. The assumption in this work is that a $200\,\mathrm{ms}$ to $300\,\mathrm{ms}$ end-to-end latency budget is sufficient. The architecture must also be causal. For the REAL-TSE baseline this meant replacing bidirectional LSTMs over time with unidirectional ones. It also included normalising over a single frame or chunk rather than over the whole sequence [3]. Claimed and measured latencies often disagreed in the challenge, so the organisers recommend reporting a component breakdown of the algorithmic latency.

A concern about streaming is that it makes it harder to identify the target speaker. As found in the survey by Žmolíková et al. [5], typical TSE systems use the entire context of an audio clip to identify the target speaker. However, in a streaming scenario, the system must identify the speaker as the audio arrives.

2.3. How extraction is evaluated

When making use of signal based metrics, practitioners are often required to use simulated data. This is because a clean reference signal is required for comparison against the model output. By nature it is quite difficult to obtain perfectly clean speech in a real-world scenario. Popular resources to use for simulated mixtures include Libri2Mix [9] and WSJ0-2mix [10]. Thus the first issue related to evaluation is the translation of performance on simulated data to real-world data. Zhang et al. [11] show that there is not necessarily a direct translation in performance: in multi-channel speech enhancement, the model with the best signal-to-distortion ratio (SDR) on simulated data more than doubled its word error rate (WER) compared with the unprocessed input on real recordings from 19.5 % to 41.5 %. Signal scores also fail in the following manner for TSE: *(i)* extracting a wrong speaker can still yield a low signal score due to similarities in speech, and *(ii)* a silent target leaves the score undefined.

A natural guess to evaluate the performance of a TSE system is to suggest using perceptual metrics. These metrics are designed to mimic what a human listener would perceive when

listening to the audio, for example DNSMOS [12]. However, as found by Wang et al. [3], these metrics can be gamed by adding adversarial perturbations to the waveform to inflate them. In other words, an extractor can be trained to maximise the perceptual metric without it improving the actual quality of the audio. Secondly, these perceptual metrics still fail in TSE as they do not identify if the wrong person is speaking.

The next progression would be to transcribe the audio and compare it against a ground truth transcription. The standard approach is to use the simple word error rate (WER) metric. However, this metric hides nuance in the failure of an extractor. Recognisers are hurt by the distortions of the extractor rather than by leftover noise left in the audio [13] and secondly, a different recogniser may have different performance on the same audio. This leads into the shortcomings of WER as a stand alone metric for TSE. Suppose a word is heard from the interfering speaker, and secondly another word that neither speaker said and was invented, both can appear in the metric as a substitution or an insertion error. Finally the WER is only defined if there is target speech present.

2.4. Speech LLMs as listeners and judges

It is not a novel idea to make use of large language models (LLMs) to evaluate the quality of speech. In fact, Manakul et al. [14] found that large audio models (LAMs), which are LLMs that accept audio as input, are able to rank speech systems similarly to human raters. However, for tasks related to measuring speaking rate or speaker identity, the performance is poor and often close to random chance. The authors also found that these established LAMs that they tested are robust against noise and are more reliable on lexical content as opposed to non-lexical content.

Using a language model as a rater is a studied problem. Zheng et al. [15] found that a strong text model can produce output that agreed with human preferences as often as humans agree with each other over 80 % of the time. However, a caveat to this is that the model's verdicts carried biases: it preferred the first option it was shown, longer answers, and output that it had generated itself. The concern is that the same biases may be present in audio models. A partial solution came from AIR-Bench [16] that scores a pair of answers twice with the order reversed to prevent positional bias which is an idea carried through in the training of the model in this work where the target and interferer swap roles in an epoch to form two trials. Secondly, a model can leverage its prior knowledge of a language to make judgements instead of actually listening to the audio. This was explored by Foo et al. [17] where audio language models keep 60–72 % of their benchmark scores even when the audio is removed. The prior was found to influence hallucination too as found by Koenecke et al. [18]. Another consideration is the handling of noisy audio. A natural response would be to enhance the audio first. However, enhancement was found to be detrimental according to Chondhekar et al. [19] where the denoising of medical speech actually raised the semantic WER of all four recognisers (including Gemini Flash 2.0) that were tested in 40 different configurations.

In Manakul et al. the study uses the model as a rater. The LAM is prompted for a verdict, such as preference over audio samples, and the final decision becomes the score. This work uses the model differently. The live model is only asked to write down what it heard and the score is computed by comparing the transcription to a ground truth text (Section 3.4.2). Each clip that is presented to the LLM is on its own, which mitigates positional bias that was found by Manakul et al. Their noise robustness was also only tested using white noise and did not explore competing speakers. Using the model as a listener rather than a rater is less common. The closest work reviewed here, Fela and Mowlaee [20], found that standard audio-quality metrics fail to predict how often speech enhancement changes an LLM's downstream output.

Therefore a gap remains. The studies making use of a model as a listener consider noise instead of competing speakers, and report a single error instead of a rate of error. None of the studies mentioned above consider the difference in error between a model leaking an interfering speaker versus inventing words that neither speaker said. This work proposes a measure to address this gap (Section 3.4.2).

3. Methodology

The methodology section describes the process and reasoning behind the data generation, extractor architecture and evaluation metrics.

3.1. Data construction

3.1.1. Why the data is constructed

There are four main considerations for the dataset that need to be met: *(i)* training requires a clean target signal as a reference for what the extractor should output, *(ii)* the data must possess exact transcriptions of the target and interferer signals to allow for accurate transcription-based evaluation, *(iii)* trials need to include examples where the target is absent, and *(iv)* the data must possess both best case and worst case scenarios in order to interpret the performance of the extractor relative to these scenarios. These considerations limit the available choices of publicly available datasets.

Constructing the data is an established practice. The 2026 REAL-TSE challenge [3] supplied no training set at all, and required that participants document whichever open-source data they used. After reviewing the submissions, the organisers describe the data recipe that the entrants converged on: synthesize mixtures from LibriSpeech, WHAM! noise and simulated room impulse responses, whilst varying ratios such as SNR, SIR, reverberation time, chunk length and enrollment quality and they inject target-absent segments. The dataset constructed here follows that recipe. The organisers further observed that several of the strongest entries were built on backbones near-identical to the official baseline architecture, the BSRNN, yet scored substantially better. The organisers attributed the success to data simulation, real-data adaptation, pseudo-label filtering, loss design and post-processing rather than to architecture

alone.

3.1.2. Source of audio

The audio data was collected from a variety of sources to produce a diverse dataset. Speech was chosen for target and interferer from the LibriSpeech [21] dataset, which is a corpus of read English speech derived from audiobooks. The background noise was chosen from WHAM! [22] which is a dataset of real-world noise recordings considering environments such as offices, streets, and cafés.

3.1.3. Trial construction

The trials aim to reflect real-world extraction scenarios. The primary focus is towards a base case of a target speaker close to a recording device with an interfering speaker in a similar proximity and faint background noise. The difficulty came in constructing the dataset to be realistic enough to simulate a real-world scenario, but also not so difficult that a simple extractor of limited capacity would be unable to learn the task. A visual representation of the trial construction can be found in Figure 3.1.

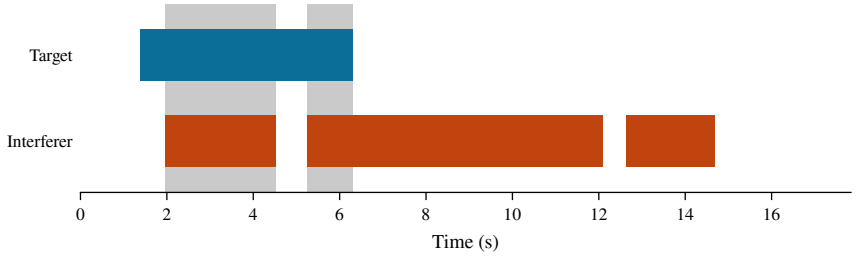


Figure 3.1: Visual representation of the trial construction. The target and interferer are drawn from LibriSpeech, and the noise bed is drawn from WHAM!. The shaded areas show the overlapping regions of speech.

A single trial consists of the following components: *(i)* target speech, *(ii)* interferer speech, *(iii)* noise bed, *(iv)* the mixture of the first three components, and *(v)* an enrollment clip of the target speaker. Each trial falls into one of four cases: *Case 1*, both speakers present, *Case 2*, the target alone, *Case 3*, the other speaker alone, and *Case 4*, noise only. The mixture is constructed to be 15–20 seconds long. The trials are drawn from two separate regimes: a base case and a difficult case. Base trials (60 %) use a slightly easier Signal-to-Noise Ratio (SNR), the reverberation time, T_{60}, is lower to simulate less echoey rooms, and ensures the target speaker talks for longer to pick up the signal better. Hard trials (40 %) use the full range of every parameter, and the Signal-to-Interference Ratio (SIR) range is the same in both. Half of the two-speaker trials pair speakers of the same gender. To simulate room acoustics, including reverberation, room dimensions, source and microphone positions, the `pyroomacoustics` [23] package is used to simulate the room impulse response. The trials are constructed to be as realistic as possible, but also to be reproducible and controllable. A sample of the room construction can be found in the Appendix A.

Enrollment. The enrollment clip is a clean $5\,\mathrm{s}$ audio segment of the target speaker from LibriSpeech and is used to provide a reference of what the target speaker sounds like. The enrollment clip is first attempted to be drawn from a different book reading. If this is unsuccessful, then the clip is drawn from a different chapter of the same book, otherwise the clip is drawn from a different utterance of the same chapter. The reason for this is to avoid the extractor memorising the target speech by learning to recognise the content rather than the *manner* that the target speaker speaks. The distribution of enrollment clips across the dataset is shown in Figure B.1 in Appendix B. Again to avoid bias, the enrollment clips are simulated to come from different recording devices. This is done by applying equalisation (EQ) augmentation to the enrollment clips by applying a random EQ filter to the clip with a probability of 0.5. This technique was suggested by the CARTSE [1] submission for the 2026 REAL-TSE challenge [3].

Parameter distribution choice. SIR is chosen uniformly from the range of -10 to 10 dB, and the SNR is chosen uniformly from the range of 0 to 20 dB. These two ratios simulate the relative loudness of the target speaker to the interferer and background noise, respectively. A negative number for the SIR indicates that the interferer is louder than the target speaker, and the lower the SNR, the louder the background noise is relative to the target speaker. The SIR was chosen to be symmetric around $0\,\mathrm{dB}$ after experimentation found that if the target was always (or the majority of the time) louder than the interferer, then the extractor would just learn to follow the speech that was loudest and hence ignore the conditioning of the enrollment audio.

Reverberation time, T_{60}, is drawn uniformly from $0.25\,\mathrm{s}$ to $0.6\,\mathrm{s}$, which covers a carpeted office to a large room with hard surfaces. The wall absorption is solved using the Sabine relation [24] rather than sampled (Appendix A). The remaining parameters are listed in Appendix B.

Target-absent trials. The extractor needs example trials to learn to rely on the conditioning of the enrollment audio to identify target speech. The target-absent trials support this idea by having the model learn to stay silent when the enrollment speaker is not present. The split is as follows for these trials: 49.5% both speakers present, 25.5% target only, 19.6% interferer only and 5.3% noise only.

Splits. The dataset is split into training, validation and test sets. Each of these sets has its own distinct set of speakers such that there is no overlap of speakers between the sets. The training set consists of 9955 trials, consisting of 1172 unique speakers. The validation set holds 200 trials from 40 other speakers: 103 with both speakers present, 47 with the target alone, 42 with the other speaker alone and 8 with noise only. It is used twice: during training, to score every epoch and choose the checkpoint (Section 4). A test set is kept apart as the final evaluation of the simulated data seen in Section 5. It holds 500 trials from 20 further speakers: 249 with both speakers present, 115 with the target alone, 109 with the other speaker alone and 27 with noise only. No model was trained, tuned or selected on it, and each model is scored on it only after its checkpoint has been chosen.

Limitations. The dataset is constructed to be as realistic as possible, but it is still a simulated dataset. The dataset is restricted to two speakers, which can confound the extractor to expect the target to be present if two speakers are present. Secondly, the combined audio forms *read* speech and not *conversational* speech. This means that the resulting audio does not have natural turn-taking, or conversational dynamics. These dynamics are simulated through the overlap of the target and interferer, but it is not a perfect simulation. Next, the dataset is restricted to English speech, and shoebox rooms only. Finally, the noise beds used are from WHAM! [22] and range in length from 5 to 20 seconds long. This means that the noise beds are not long enough occasionally to cover the entire trial. To solve this, the noise bed is looped to cover the entire trial.

3.1.4. Real-recording evaluation set

The constructed dataset is limited to simulated rooms and read speech. To provide a final evaluation of the extractor, the AMI corpus [25] is used. The AMI corpus contains 100 hours of real recordings of meetings with multiple speakers. Trials are kept to 10–20 seconds and are cut following the method of REAL-T [26]. The extractor is required to listen to a single table microphone and the enrollment audio is of the target speaker speaking alone into their headset from a different meeting.

A difficulty is that the AMI corpus does not have clean reference signals or entirely verbatim transcriptions. Another difficulty is that the enrollment is from a headset which can pick up other speakers and background noise, thus the enrollment becomes an approximation rather than a perfect one. Finally, the mixture is far quieter than in the constructed dataset where the median level of the target is $-48.6\,\mathrm{dB}$ compared to the validation set of $-29.2\,\mathrm{dB}$.

Final evaluation split. As a final test, the extractor is tested on real recordings using the AMI corpus. The AMI evaluation contains 300 trials from 58 meetings, totalling 77.9 minutes. There are 60 speakers and all trials are Case 1 (both speakers present). The full summary can be found in Appendix C.

3.2. Baseline model architecture

The baseline architecture section describes the architecture that was used as a backbone in the REAL-TSE baseline [3]. The architecture in this work is built from scratch, but the design is that of the baseline.

3.2.1. Band-split Recurrent Neural Network (BSRNN)

Luo and Yu [7] propose the BSRNN model as a frequency-domain separation model that is designed for high sample rate audio. The implemented BSRNN is adapted to speech with a Time-Frequency Map enrollment conditioning module [27] to turn it into a target speaker extraction model.

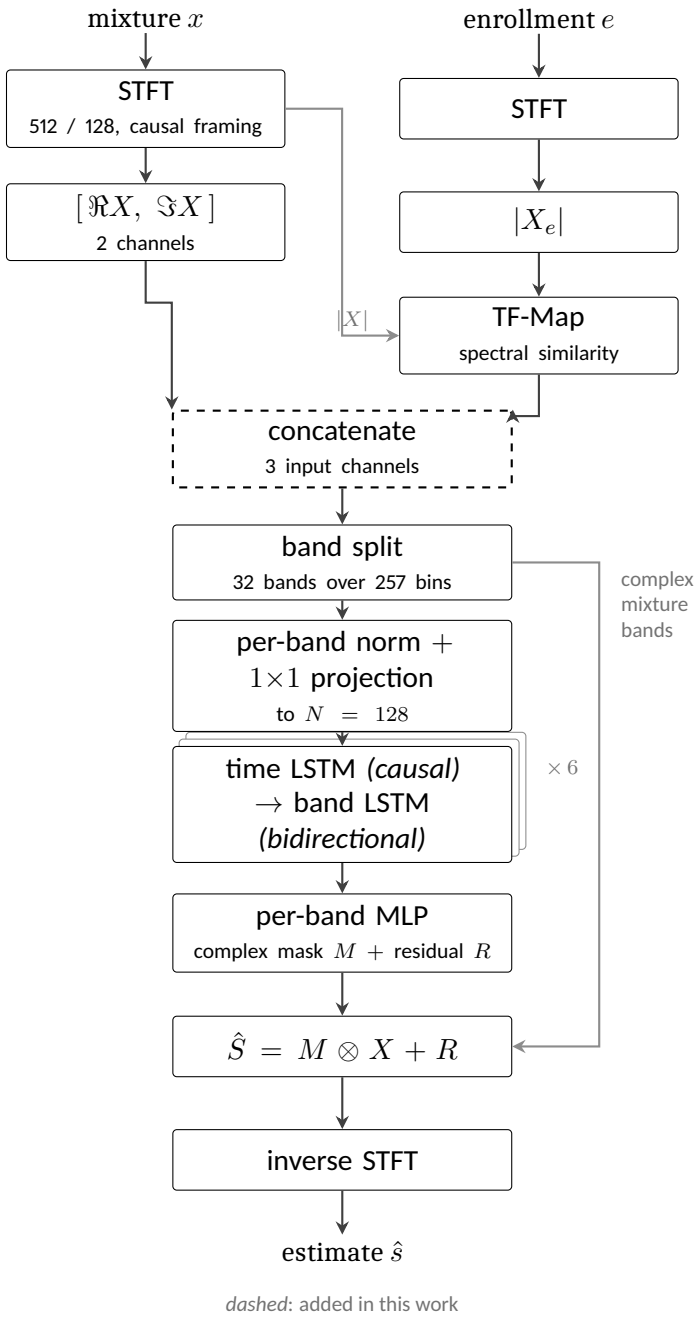


Figure 3.2: The extractor as implemented. Band splitting and the band-level and sequence-level modelling follow [7], the mask-plus-residual estimator follows [8] and the TF-Map conditioning follows [27] [AI3].

3.2.2. Short-time Fourier transform

The model operates in the time-frequency domain rather than a waveform. To ensure the input audio is in suitable form, a short-time Fourier transform (STFT) is applied by cutting the waveform into short, overlapping frames. A window function is applied to each frame, and then a discrete Fourier transform is applied to each windowed frame. The result is a complex-valued spectrogram, a two-dimensional representation of the audio signal. On the horizontal axis is

time, and on the vertical axis is frequency. The discrete Fourier transform is defined as:

$$X(f,t) = \sum_{n=0}^{N-1} x[tH+n]\, h[n]\, e^{-\mathrm{j}2\pi f n/N}, \tag{3.1}$$

where N is the window length, H the hop between frames, $h[n]$ a Hann window and j the imaginary unit. Following Yu et al. [8], $N = 512$ and $H = 128$ at $16\,\mathrm{kHz}$: a $32\,\mathrm{ms}$ window moving in $8\,\mathrm{ms}$ steps, giving 257 frequency bins. Frames are not centred on timestamps as this would imply future information which is not suitable for streaming tasks. Instead, the signal is padded with $N - H$ samples at the start of a sequence.

3.2.3. Band plan

Figure 3.3 shows the band plan against a real mixture in the training set. The band plan in this work follows the implementation from Zhang et al. [27] for their $16\,\mathrm{kHz}$ target speaker extraction model and used by the challenge baseline built on it. The split is 257 frequency bins into 32 bands as seen in the figure, but no reason is given in the original paper for these particular boundaries, and no explanation is given in previous papers. More bands are allocated to the lower frequencies as speech is more densely concentrated there and requires more resolution.

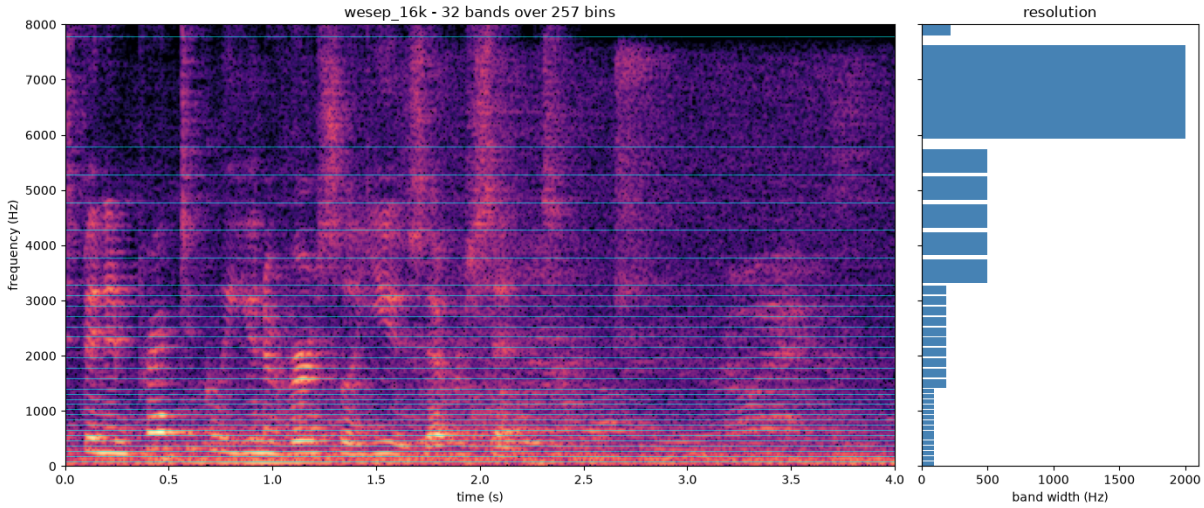


Figure 3.3: The band plan used here, shown against a four-second mixture from the training set. The right graph shows the width of each band in hertz.

3.2.4. Per-band normalisation

Normalisation is applied across the values of a band *within a single frame*. This is a deliberate design choice that *opposes* Yu et al. [8], who use batch normalisation for their online configuration. The opposition is for the following reasons: *(i)* during training, the mean and variance are computed from the current batch, but during inference, these statistics are computed from the entire training set. The function that is optimised during training is therefore not the function that is deployed. *(ii)* The variability of the data does not give a reliable estimate of the true

distribution. The statistics of a batch therefore depend on what was drawn where trials that will not reflect the statistics seen at inference. *(iii)* Single frame normalisation carries no state and is unaffected by the amount of audio processed, so a four-second training chunk is normalised exactly as a stream that has been running for a minute.

3.2.5. Residual LSTM block

Two kinds of sequential structures are modelled: time and frequency. Time is unidirectional, but frequency is bidirectional, as all of the bands of a frame are available at the same instant and can be traversed in either direction. The block is used for both kinds of modelling, with the same structure but different parameters. At inference time, since the model is running in a streaming fashion, the LSTM's internal state is preserved between frames and information can be carried forward indefinitely. The LSTM is therefore a natural fit for the task, and it is used in the same way for both the time and frequency sequences. Each block completed the following taking inspiration from [7]: the input is normalised, passed through the LSTM, and projected back to the original width. Finally, the input is added back to the output of the projection, forming a residual connection. A single LSTM layer is given a width of 192, which is Yu et al.'s [8] choice for their $16\,\mathrm{kHz}$ model. The input to the block is 128 wide, which is projected to 192 for the LSTM and then projected back to 128 for the residual addition.

3.2.6. Band and sequence model

The band and sequence model is the implementation of the residual LSTM block of Section 3.2.5 in a stack of six blocks again following Yu et al. [8], using the input from Section 3.2.4. After the per-band normalisation and projection, there are 32 bands of 128 features each. Each of these blocks make two passes: over the time dimension (unidirectional), and then make a pass over the frequency dimension (bidirectional). Pairing the two passes in a single block is the design of Luo and Yu [7], who motivate it by analogy with dual-path recurrent networks for speech separation rather than by an argument specific to bands. The effect of stacking the blocks is to allow for the accumulation of evidence over longer time spans.

3.2.7. Estimation module

The estimation module is responsible for producing an output spectrogram that contains the estimated target speaker's speech. This is done by creating an output network per band to reconstruct the target speaker's speech from the features produced by the band and sequence model. Yu et al. realised it would be more efficient to mask the mixture rather than predict the output spectrogram directly. This mask is a number for each time-frequency bin, which the mixture is multiplied by. A mask alone is not enough, as multiplication can only ever scale what a bin already holds. In the cases where the interferer completely dominates a bin, no multiplier recovers the target, as the multiplication operation would not be able to scale the drowned out target. To solve this, the estimation module makes use of a residual spectrogram, which is predicted directly and added. Yu et al. [8] introduce it for exactly this reason, to absorb what the

mask alone produces as artefacts. It is useful to not bound this mask and is what the original implementation encouraged. The estimate for a band is then

$$\hat{S} = M \otimes X + R, \tag{3.2}$$

where $\otimes$ is the complex product, taken bin by bin. The X that the mask multiplies is the original complex mixture and is taken as is from the band split and not from the output of the model's features. The features are used only to *decide* how to mask the input. The conditioning feature of Section 3.2.8 therefore never enters the signal path either.

3.2.8. Time-frequency map

The TF Map [27] is the module that turns speech enhancement into target speaker extraction. It compares the mixture with the clean enrollment clip frame by frame, and gives the network a cue of what the target's spectrum should look like at each instant. Let $|X_t|$ be frame t of the mixture's magnitude spectrogram and $|X_{e,j}|$ frame j of the enrollment's. Every frame is first scaled to unit length, so that only the spectral *shape* remains and does not keep the loudness:

$$\tilde{x}_t = \frac{|X_t|}{\big\||X_t|\big\|}, \qquad \tilde{e}_j = \frac{|X_{e,j}|}{\big\||X_{e,j}|\big\|}. \tag{3.3}$$

Each mixture frame is then compared with every enrollment frame by cosine similarity. For unit vectors this is simply the inner product. The similarities are then turned into weights that blend the enrollment frames into a template that can be referenced against the mixture frame:

$$C_{t,j} = \big\langle \tilde{x}_t,\ \tilde{e}_j \big\rangle, \qquad A_{t,j} = \operatorname*{softmax}_j \big(\kappa\, C_{t,j}\big), \qquad \mu_t = \sum_{j=1}^{T_e} A_{t,j}\, \tilde{e}_j. \tag{3.4}$$

Zhang et al. [27] also normalise the frames and make use of cosine similarity, but they use softmax directly on the inner product rather than scaling it by a factor $\kappa = 16$. After testing it was found that since the similarities lie between $[0, 1]$, the softmax spread the weights too evenly and thus did not give a clear template. The κ factor is therefore used to scale the similarities which sharpens the softmax to give a more distinct template. The template only describes the shape and does not indicate whether a target is speaking. As in [27], the energy is restored by projecting the mixture frame onto the template's direction:

$$u_t = \frac{\mu_t}{\|\mu_t\|}, \qquad m_t = \big\langle\, |X_t|,\ u_t \,\big\rangle, \qquad \phi_t = m_t\, u_t, \tag{3.5}$$

where m_t is a scalar per frame which describes how much of the mixture's energy lies along the target's predicted spectrum. ϕ_t, the template scaled by it, is the cue passed to the network. The TF Map itself has nothing to train as (3.3)–(3.5) contain no weights. The extra channel is the only addition as it widens the per-band projection by about $33\,000$ parameters.

3.2.9. Objective function

The objective has to express two different requirements: output the target signal when it is present, otherwise, output silence. A single loss cannot serve both use cases and so the objective is split, and each crop is scored by the branch (target present versus absent) that applies to it.

Throughout, x is the mixture, s the target reference and $\hat{s}$ the model's output, all in the time domain over one training crop.

Target present. The first term measures how much of the output is the target and how much is everything else, following [1]. The loss needs to be invariant to the overall volume: if $\hat{s}$ has the correct shape, it should not be penalised for being loud or quiet. Therefore the first step is to find the single gain that best explains the output as a scaled copy of the target, which will be calculated as a projection,

$$\alpha = \frac{\langle \hat{s},\ s \rangle}{\|s\|^2}, \qquad s_{\text{proj}} = \alpha s, \tag{3.6}$$

and then to compare the energy of that scaled target against the energy of whatever the output contains beyond it. The loss is modelled after the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR). The loss is therefore,

$$L_{\text{pres}} = -10 \log_{10} \frac{\|s_{\text{proj}}\|^2}{\|\hat{s} - s_{\text{proj}}\|^2 + \tau_{\text{pres}} \|s_{\text{proj}}\|^2}. \tag{3.7}$$

Lower is better. The $\tau_{\text{pres}} = 10^{-3}$ term in the denominator is a floor and caps the best score at $-30\,\text{dB}$, which is a suggestion from CARTSE [1]. An output whose residual error already sits $30\,\text{dB}$ below the target is close enough to be treated as solved, so it would be wasteful to keep spending gradient on it. The measure has two blind spots. It views the leaked interferer, noise and artefacts as an identical error per unit of energy. Secondly, the ratio is dominated by the low frequencies and thus the quieter bands that carry the consonants can be reconstructed poorly without much penalty.

Multi-resolution spectral term. Equation (3.7) cannot express *where* in the spectrum the error lies, a model that reconstructs the loud low bands well and the quiet high bands poorly is barely penalised and, being scale-invariant, it does not constrain the output's absolute loudness at all. Thus the to address this, we consider multiple resolutions of s and $\hat{s}$ such that the reconstruction of both the low and high frequency bands at different scales is encouraged. A spectral term allows for this following [8]:

$$L_{\text{MR}} = \frac{1}{|\mathcal{N}|} \sum_{N \in \mathcal{N}} \Big(\big\| \, |S_N|^p - |\hat{S}_N|^p \, \big\|_1 + \big\| \Re S_N - \Re \hat{S}_N \big\|_1 + \big\| \Im S_N - \Im \hat{S}_N \big\|_1 \Big), \tag{3.8}$$

where S_N and $\hat{S}_N$ are the spectrograms of the target and the output (Equation 3.1) with window length N and hop $N/4$, and each norm is averaged over its time-frequency coefficients.

The exponent $p = 0.3$ compresses magnitudes, which will highlight the quieter components

relative to loud ones so that an error in a low-energy region is not diluted just because it is quiet. Without the compression the term would be dominated by the same loud low bands that already dominate (3.7). The real and imaginary terms are left uncompressed and they are what make the term sensitive to phase: a magnitude-only loss would be indifferent to the phase that the complex mask of Section 3.2.7 is free to change. Finally the term is evaluated at several window lengths namely: 8, 16, 32 and $64\,\mathrm{ms}$. This is with respect to the model's own $32\,\mathrm{ms}$ framing. The model builds its output by masking $32\,\mathrm{ms}$ frames and overlapping them, so the places where most errors would occur can be expected to exist at frame boundaries. The $8\,\mathrm{ms}$ and $16\,\mathrm{ms}$ windows resolve this and the $64\,\mathrm{ms}$ window catches harmonic structure that the short ones blur.

Target absent. The model should penalise any output that is not silence. This is modelled after a modified version of the absolute level loss from CARTSE [1], where in this work the loss now contains the denominator term.

$$L_{\text{abs}} = 10\log_{10}\frac{\|\hat{s}\|^2 + \tau_{\text{abs}}\|x\|^2}{\|x\|^2}. \tag{3.9}$$

Including the denominator term makes the loss relative to the input, which is directly readable: $0\,\mathrm{dB}$ means the output is as loud as the mixture, that is, the model did nothing; $-10\,\mathrm{dB}$ means the interfering speech was suppressed by $10\,\mathrm{dB}$. The same τ_{abs} floors the term at $-20\,\mathrm{dB}$, beyond which the output counts as silent. A positive value means the model amplified the mixture, which is a direct failure. The inclusion of the denominator also allows for scale invariance, which puts the loss in the same units as the loss for the target present case.

Output level. L_{pres} is scale-invariant and L_{abs} rewards silence, leaving a gap in the training that would otherwise allow a model to simply mute its output. This gap is filled by the third term, which ensures that the model does the following: *(i)* prevents the indiscriminate output of silence, and *(ii)* ensures a reasonable level of output. It bounds the output level against the target's, with a dead zone of $\pm\delta$:

$$L_{\text{gain}} = \max\!\left(0,\ \left|20\log_{10}\frac{\mathrm{rms}(\hat{s})}{\mathrm{rms}(s)}\right| - \delta\right), \qquad \delta = 3\,\mathrm{dB}. \tag{3.10}$$

Inside $\pm 3\,\mathrm{dB}$ of the target's level the term is exactly zero, where rms is the root mean square. Its weight, $w_{\text{g}} = 1.69$, is derived rather than chosen: it is the geometric midpoint of the weight at which muting stops paying (where what muting saves on (3.9) equals what it costs on (3.10), both weighted as in (3.11)) and the weight where output level will outweigh the actual content extracted.

Combining the two branches. Each term is averaged over the crops it applies to, and the two averages are weighted against one another:

$$L = (1-w)\underset{\text{target present}}{\text{mean}}\Big[\,L_{\text{pres}} + w_{\text{m}}L_{\text{MR}} + w_{\text{g}}L_{\text{gain}}\,\Big] + w\underset{\text{target absent}}{\text{mean}}\Big[\,L_{\text{abs}}\,\Big]. \tag{3.11}$$

The decision to average over the crops within each branch rather than over the entire batch is deliberate. During an epoch, a random crop can draw a trial whose label says both target and interferer are present, but the crop itself may contain silence (occurs in $5.8\,\%$ of cases). The model should therefore be rewarded for silence where the loss should not be diluted by the presence of other crops in the batch. Therefore the branch a crop belongs to is decided from the audio of the cropped target itself, not from the trial's label.

Weights. The weight w on the silent branch is derived rather than chosen. Silent crops make up a measured $p = 0.297$ of the training crops and, following [1], each silent crop is counted twice ($\eta = 2$). The present branch then carries $1-p = 0.703$ and the absent branch $\eta p = 0.594$, and normalising the two to sum to one gives

$$w = \frac{\eta p}{(1-p) + \eta p} = \frac{0.594}{1.297} = 0.458.$$

The weight w_{m} on the spectral term is derived from the target present and absent losses. The reason for this term is that equations (3.7) and (3.8) are in different units. L_{pres} is in units with respect to a ratio in decibels, and L_{MR} is the mean absolute difference between compressed magnitudes. Therefore the weighting term w_{m} will reconcile the two terms to be on the same scale. This is done by evaluating on the output that does nothing at all, $\hat{s} = x$, over 300 training crops which will simulate the scenario for the average losses for an untrained model. The median values for L_{pres} and L_{MR} are -5.91 and 0.184 respectively. The aim is for the spectral term (Multi-Resolution) to be roughly $30\,\%$ of the present branch at the start of training. The calculation then becomes

$$w_{\mathrm{m}} = \frac{0.3 \times |L_{\text{pres}}|}{L_{\text{MR}}} = \frac{0.3 \times 5.9088}{0.1842} = 9.62$$

3.3. Extension model architecture

The extension architecture is based on the baseline architecture of Section 3.2, with two modifications. The first gives the network more information about the target speaker's voice (Section 3.3.2), together with a change to how the cue is passed to the network (Section 3.3.1). The second adds a gate to better infer whether the target speaker is present in a frame (Section 3.3.3).

3.3.1. Splitting the cue into parts

The baseline architecture passes the network the TF-Map cue of [27] and includes the energy-recovery projection (Equation 3.5). By Equation 3.3, every mixture frame is its loudness times its shape, $|X_t| = \left\||X_t|\right\| \tilde{x}_t$. Substituting this into m_t of Equation 3.5 shows what the baseline's single number contains:

$$m_t = \left\langle\, |X_t|,\ u_t \,\right\rangle = \left\||X_t|\right\| \left\langle \tilde{x}_t,\ u_t \right\rangle = \left\||X_t|\right\| c_t, \tag{3.12}$$

where

$$c_t = \cos\theta_t = \left\langle \tilde{x}_t,\ u_t \right\rangle \tag{3.13}$$

is the cosine similarity of Equation 3.4 which is measured with the blended template of u_t instead of an enrollment frame. The problem with m_t in the baseline architecture is that once multiplying the loudness and the similarity into a scalar, the network cannot distinguish between a frame that is very loud but not similar to the target and a frame that is very similar but not loud. The idea is to pass the parts instead of the product [AI6a]. Two parts are already defined: the template direction u_t of Equation 3.5 and the similarity c_t of Equation 3.13. The third part is new and is the residual

$$r_t = |X_t| - m_t u_t \tag{3.14}$$

which is the part of the mixture frame that the template of the target does not explain. A property of the residual is that it is orthogonal to the template, shown by:

$$\left\langle r_t, u_t \right\rangle = \left\langle |X_t| - m_t u_t, u_t \right\rangle = \left\langle |X_t|, u_t \right\rangle - m_t \left\langle u_t, u_t \right\rangle = m_t - m_t = 0. \tag{3.15}$$

where the last equality uses the fact that u_t is a unit vector. The orthogonality thus allows a frame to be split exactly into the equation:

$$|X_t| = m_t u_t + r_t \tag{3.16}$$

Therefore we can rebuild ϕ_t exactly as:

$$\phi_t = c_t \left\| |X_t| \right\| u_t \tag{3.17}$$

3.3.2. Speaker context embedding

When considering Section 3.3.1 the cue is still only described by the enrollment's spectra. Target speakers that have similar pitch and vocal characteristics thus produce similar templates. Measured on the baseline architecture, the extractor was able to select the right voice $71.8\,\%$ of the time when two speakers were of different gender, but only $52.6\,\%$ when they were the same gender. The idea for this extension is to incorporate further target speaker identity. The second extension is to add a speaker embedding for each target speaker. Each enrollment clip is passed through a frozen speaker recognition network, ECAPA-TDNN [28] using the SpeechBrain checkpoint trained on VoxCeleb [29], to produce a 192-dimensional embedding. The embedding is then passed through a small learned layer to produce a 128-dimensional vector to be used in the extractor. ECAPA itself is large at 20.77 million parameters and hence would cause storage concerns if it were included in the extractor. However, the ECAPA network can be run once per enrollment clip and the resulting embedding can be stored for later use. This allows for the extractor to be served locally on a device without needing to have the full

ECAPA network, only the stored embedding for the target speaker.

The WeSep toolkit [30], an established open-source implementation of the baseline architecture, supports two uses of the embedding. The first (which is the one used in this work) is to consider utterance-level embedding, and the second is to consider contextual embedding of Zhang et al. [27]. Although WeSep supports the contextual embedding, it was discarded as an option for this work because it recomputes the embedding for every frame of the mixture, which is computationally expensive.

3.3.3. Target presence gate

The baseline architecture has no explicit, underlying mechanism to determine whether the target speaker is present in a frame. The gate provides this probabilistic estimate and is trained jointly with the extractor. The gate represents a two-state Markov chain, where the states are *present* and *absent*.

Gated output. Let $Z_t \in \{0, 1\}$ be the random variable that is 1 if the target speaker is present in frame t and 0 otherwise. The extractor's output frame from (3.2) is then gated by whether the target speaker is present by z_t:

$$\hat{S}_t(z_t) = z_t\Big(M_t \otimes X_t + R_t\Big), \tag{3.18}$$

The gate reads the separator output $u_{k,t} \in \mathbb{R}^{128}$ of band $k \in \{1, \dots, 32\}$ (Section 3.2.6):

$$h_t = \frac{1}{32}\sum_{k=1}^{32} u_{k,t}, \qquad \hat{h}_{t,n} = \frac{h_{t,n} - \mu_t}{\sqrt{s_t^2 + \epsilon}}, \qquad \pi_t = \sigma(w^\top \hat{h}_t + b), \tag{3.19}$$

with $\mu_t = \frac{1}{128}\sum_{n=1}^{128} h_{t,n}$, $s_t^2 = \frac{1}{128}\sum_{n=1}^{128}(h_{t,n} - \mu_t)^2$ and $\epsilon = 10^{-5}$. Here h_t is the average of the bands state, $\hat{h}_t$ is that state standardised across the channels. π_t is the probability that the target speaker is present in frame t, $w \in \mathbb{R}^{128}$, $b \in \mathbb{R}$, and let $\phi = (w, b)$. The approximate posterior over the sequence of target presence variables $z_{1:T}$ [31] is then given by:

$$q_\phi(z_{1:T} \mid x, e) = \prod_{t=1}^{T} \mathrm{Bern}(z_t; \pi_t), \tag{3.20}$$

where x is the mixture and e is the enrollment. The product over frames follows the mean field assumption.

Markov Prior. The prior leverages the first order Markov property (as a target who speaks in one $8\,\mathrm{ms}$ frame is likely to speak in the next [32]) and is factorised as:

$$\begin{aligned} p(z_{1:T}) &= p(z_1)\prod_{t=2}^{T} p(z_t \mid z_{t-1}) \\ &= p(z_1)\prod_{t=2}^{T} A_{z_{t-1}, z_t} \quad \text{where } A_{i,j} = p(z_j \mid z_i) \end{aligned} \tag{3.21}$$

A is fixed from the training labels and is the transition matrix of the Markov chain. Let $D = d$ denote the random variable for the number of consecutive frames of a speech run (consecutive $z_t = 1$). Therefore, expressed as a probablity:

$$\begin{aligned} p(D = d) &= A_{1,1}^{d-1} A_{1,0}, \qquad d \in \{1, 2, \dots\} \\ &= A_{1,1}^{d-1}(1 - A_{1,1}) \end{aligned} \tag{3.22}$$

therefore D is a geometric distribution with

$$\mathbb{E}[D] = \frac{1}{1 - A_{1,1}}. \tag{3.23}$$

Next, to set the transition matrix A, sample the mean speech run length in the training data, $\bar{d}_{\text{on}}$, which is the maximum likelihood estimate of $\mathbb{E}[D]$, and treating silence runs (sample mean $\bar{d}_{\text{off}}$) the same way, gives

$$A_{11} = 1 - \frac{1}{\bar{d}_{\text{on}}}, \quad A_{10} = \frac{1}{\bar{d}_{\text{on}}}, \quad A_{00} = 1 - \frac{1}{\bar{d}_{\text{off}}}, \quad A_{01} = \frac{1}{\bar{d}_{\text{off}}}. \tag{3.24}$$

Frame 1 has no previous frame, so it takes the stationary probability ρ of the chain. By the law of total probability, with $P(z_t = 1) = P(z_{t-1} = 1) = \rho$,

$$\rho = \rho A_{11} + (1 - \rho) A_{01} \implies \rho = \frac{A_{01}}{A_{01} + A_{10}} \tag{3.25}$$

The actual transition matrix used in the prior can be found in Appendix D.

Objective. Training should maake the clean target speech s as probable as possible given the mixture and the enrollment.

$$\begin{aligned} \log p_\theta(s \mid x, e) &= \log \sum_z p_\theta(s \mid x, e, z)\, p(z) \\ &= \log \mathbb{E}_{z \sim q_\phi(z \mid x, e)} \left[\frac{p_\theta(s \mid x, e, z)\, p(z)}{q_\phi(z \mid x, e)} \right] \\ &\geq \mathbb{E}_{z \sim q_\phi(z \mid x, e)} \left[\log \frac{p_\theta(s \mid x, e, z)\, p(z)}{q_\phi(z \mid x, e)} \right] \quad \text{By Jensen's Inequality} \\ &= \mathbb{E}_{z \sim q_\phi(z \mid x, e)} \Big[\log p_\theta(s \mid x, e, z) \Big] - \mathbb{E}_{z \sim q_\phi(z \mid x, e)} \Big[\log q_\phi(z \mid x, e) - \log p(z) \Big] \\ &= \mathbb{E}_{z \sim q_\phi(z \mid x, e)} \Big[\log p_\theta(s \mid x, e, z) \Big] - \mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z) \Big). \end{aligned} \tag{3.26}$$

The first term of (3.26) requires a probability, but only a loss is available during training: $\ell(\hat{s}, s) = (\hat{s} - s)^2$, the loss of (3.11) on the gated output. Let the loss be modelled as a Gaussian

with $\kappa = 2\sigma^2$:

$$p_\theta(s \mid x, e, z) = \frac{1}{Z} \exp\left(-\frac{\ell(\hat{s}(z), s)}{\kappa}\right), \qquad \log p_\theta(s \mid x, e, z) = -\frac{\ell(\hat{s}(z), s)}{\kappa} - \log Z \tag{3.27}$$

where Z is the normalisation constant. Substitute (3.27) into (3.26) to give:

$$\begin{aligned} \log p_\theta(s \mid x, e) &\geq \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[-\frac{\ell(\hat{s}(z), s)}{\kappa} - \log Z\Big] - \mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z)\Big) \\ &= -\frac{1}{\kappa}\mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\ell(\hat{s}(z), s)\Big] - \log Z - \mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z)\Big) \end{aligned} \tag{3.28}$$

Let the right hand side of (3.28) form part of the objective to obtain

$$\begin{aligned} \mathcal{L}(\theta, \phi) &= -\frac{1}{\kappa}\,\mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\ell(\hat{s}(z), s)\Big] - \log Z - \mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z)\Big) \\ -\kappa\,\mathcal{L}(\theta, \phi) &= \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\ell(\hat{s}(z), s)\Big] + \kappa \log Z + \kappa\,\mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z)\Big) \\ &\propto \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\ell(\hat{s}(z), s)\Big] + \kappa\,\mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z)\Big) \end{aligned} \tag{3.29}$$

The ELBO at this point does not use the labels v_t for whether the target is present and thus nothing forces π_t to follow whether the target is speaking. Since v_t is known for every frame in training, it was decided to be included in a binary cross entropy between π_t and v_t. This idea was suggested by Kingma et al. [33] in an image classification context. The final objective to train the presence gate is then:

$$\mathcal{J}(\theta, \phi) = \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\ell(\hat{s}(z), s)\Big] + \kappa\,\mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z)\Big) - \lambda \sum_{t=1}^{T}\Big[v_t \log \pi_t + (1-v_t)\log(1-\pi_t)\Big] \tag{3.30}$$

Closed Form KL. The KL term of (3.29) is intractable. Suppose

$$\mathrm{KL}\Big(q_\phi(z \mid x, e) \,\|\, p(z)\Big) = \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\log q_\phi(z \mid x, e)\Big] - \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\log p(z)\Big] \tag{3.31}$$

then term 1 can be simplified as follows:

$$\begin{aligned} \log q_\phi(z \mid x, e) &= \sum_{t=1}^{T}\Big[z_t \log \pi_t + (1 - z_t)\log(1 - \pi_t)\Big] \\ \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\log q_\phi(z \mid x, e)\Big] &= \sum_{t=1}^{T}\Big[\pi_t \log \pi_t + (1 - \pi_t)\log(1 - \pi_t)\Big] \end{aligned} \tag{3.32}$$

where $\mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[z_t\Big] = \pi_t$. Term 2 simplifies as follows:

$$\log p(z) = \log p(z_1) + \sum_{t=2}^{T} \log A_{z_{t-1} z_t} \tag{3.33}$$

Recalling that being in state i at a time frame t is modelled by the Bernoulli distribution, therefore for the first frame only consider the case of belonging to a state with no transitions, and from the first state onwards, consider the transitions. The expectation of (3.33) is then:

$$\begin{aligned}
\mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\log p(z)\Big] &= \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[\log p(z_1) + \sum_{t=2}^{T} \log A_{z_{t-1}z_t}\Big] \\
&= \mathbb{E}_{z\sim q_\phi(z|x,e)}\Big[z_1 \log\rho + (1-z_1)\log(1-\rho) \\
&\quad + \sum_{t=2}^{T}\Big(z_{t-1}z_t \log A_{11} + z_{t-1}(1-z_t)\log A_{10} \\
&\quad + (1-z_{t-1})z_t \log A_{01} + (1-z_{t-1})(1-z_t)\log A_{00}\Big)\Big]^{\dagger} \\
&= \pi_1 \log\rho + (1-\pi_1)\log(1-\rho) + \\
&\quad \sum_{t=2}^{T}\Big[\pi_{t-1}\pi_t \log A_{11} + \pi_{t-1}(1-\pi_t)\log A_{10} \\
&\quad + (1-\pi_{t-1})\pi_t \log A_{01} + (1-\pi_{t-1})(1-\pi_t)\log A_{00}\Big]^{\ddagger}
\end{aligned} \tag{3.34}$$

† where $p(z_1 = 1) = \rho$ from (3.25), and since $z_t \in \{0,1\}$ exactly one of the four products is 1 for each pair (z_{t-1}, z_t), which selects the matching transition. ‡ where $\mathbb{E}_z\Big[z_{t-1}z_t\Big] = \mathbb{E}_z\Big[z_{t-1}\Big]\mathbb{E}_z\Big[z_t\Big] = \pi_{t-1}\pi_t$ by the mean field assumption, following the completely factorised variational inference of Ghahramani and Jordan [34]. Therefore subtracting (3.34) from (3.32) and removing $t = 1$ from the first summation yields

$$\begin{aligned}
\mathrm{KL}\Big(q_\phi(z \mid x, e)\,\|\,p(z)\Big) &= \pi_1\Big(\log\frac{\pi_1}{\rho}\Big) + (1-\pi_1)\Big(\log\frac{1-\pi_1}{1-\rho}\Big) + \\
&\quad \sum_{t=2}^{T}\Big[\pi_{t-1}\Big[\pi_t \log\frac{\pi_t}{A_{11}} + (1-\pi_t)\log\frac{1-\pi_t}{A_{10}}\Big] + \\
&\quad (1-\pi_{t-1})\Big[\pi_t \log\frac{\pi_t}{A_{01}} + (1-\pi_t)\log\frac{1-\pi_t}{A_{00}}\Big]\Big]
\end{aligned} \tag{3.35}$$

Next using the identity of $\mathrm{KL}\Big(\mathrm{Bern}(a)\,\|\,\mathrm{Bern}(b)\Big) = a\log\frac{a}{b} + (1-a)\log\frac{1-a}{1-b}$ we can simplify the KL term to a closed form:

$$\begin{aligned}
&= \mathrm{KL}\Big(\mathrm{Bern}(\pi_1)\,\|\,\mathrm{Bern}(\rho)\Big) + \sum_{t=2}^{T}\Big[\pi_{t-1}\,\mathrm{KL}\Big(\mathrm{Bern}(\pi_t)\,\|\,\mathrm{Bern}(A_{11})\Big) + \\
&\quad (1-\pi_{t-1})\,\mathrm{KL}\Big(\mathrm{Bern}(\pi_t)\,\|\,\mathrm{Bern}(A_{01})\Big)\Big]
\end{aligned} \tag{3.36}$$

Training and inference. The expectation of $\ell(\hat{s}(z), s)$ of Equation 3.30 will be too expensive to compute as an average over all 2^T possible sequences of $z_{1:T}$. Instead leverage Monte Carlo sampling to approximate the expectation [AI6b]. Only one sequence of $z_{1:T}$ is drawn per training clip and is an unbiased estimate of the expectation [31]. A sample $z_t \in \{0,1\}$ drawn from a

$\mathrm{Bern}(\pi_t)$, however, will potentially not change when π_t is changed slightly, so $\partial z_t/\partial \pi_t = 0$ and thus the extraction loss sends no gradient to the gate. This is solved using a soft draw with the binary Concrete distribution [35, 36].

$$\tilde{z}_t = \sigma\Big(\frac{1}{\tau}\Big(\log\frac{\pi_t}{1-\pi_t} + \log U_t - \log(1-U_t)\Big)\Big), \qquad U_t \sim \mathcal{U}(0,1), \tag{3.37}$$

where $\tau > 0$ is a temperature. The noise U_t does not depend on ϕ, so $\tilde{z}_t$ is a differentiable function of π_t (the reparameterisation trick [31]), and as $\tau \to 0$ it tends to a hard $\mathrm{Bern}(\pi_t)$ draw where τ is annealed, linearly per epoch from 1 to 0.5. At inference a single deterministic output is needed. The output that minimises the expected squared error to the random gated output (3.18) is its mean,

$$\hat{S}_t = \mathbb{E}_{z_t \sim \mathrm{Bern}(\pi_t)}\Big[z_t\,(M_t \otimes X_t + R_t)\Big] = \pi_t\,(M_t \otimes X_t + R_t), \tag{3.38}$$

so at test time each frame is scaled by π_t and nothing is sampled.

3.4. Metrics

The metrics are where the main contribution of the work lies. Since the main task is intelligibility of the target speaker for live transcribers and not necessarily the *quality* of the extracted target speaker speech, the metrics that are designed in this section aim to reflect that objective.

3.4.1. Transcripts and references

When a system produces output, either the offline or the live model with the prompt of Section E.1, it gives a transcript. The two anchors get transcribed as well to give the untouched mixture and the clean target. Every transcript is scored against the LibriSpeech reference transcript of the target speaker, and the interferer speaker. Before scoring, each transcript is normalised using Whisper's `EnglishTextNormalizer` [37, 38]. This includes lowercasing, removing punctuation, contractions expanded and numbers written consistently.

3.4.2. Live-model Content Fidelity Word Error Rate

The Live-model Content Fidelity Word Error Rate (LCF-WER) a word error rate applied to a live-model transcription of the audio. The equation is as follows:

$$\mathrm{WER} = \frac{S + D + I}{N} \tag{3.39}$$

where S is the number of substitutions, D is the number of deletions, I is the number of insertions, and N is the total number of words in the reference. It is essentially a minimum edit-distance formula, where the number of edits required to change a reference text into the proposed text is divided by the number of words in the reference text. Metric can go above 100 %.

3.4.3. Interferer Content Rate

The Interferer Content Rate (ICR) is the first metric used to highlight one particular nuance of the LCF-WER metric: when the live model transcribes a word that the target never said, that word was either invented by the model or heard from the interfering speaker. The Interferer Content Rate (ICR) measures the second case [AI7].

Content Words. The ICR metric needs to identify which words are likely to be words that are not filler words that a model can easily hallucinate. Firstly, the words that are not considered content words are removed from consideration using a stopword list, taken from the NLTK English stopword corpus [39]. Function words are removed because almost any two English sentences share them, so an overlap counted over these words would rather count as grammatical issues than content. These content words will be used to make an interferer-exclusive bag of words, which indicates which words the model picks up from the interfering speaker. Specifically, a content word is *available to leak* on trial i if the interfering speaker said it and the target did not. Each word is counted once per trial: a_i is the number of words available to leak on trial i, and ℓ_i the number of those that leaked.

The rate. The number of leaked words grows with how much the interfering speaker said, so the count is turned into two rates. The first asks how often a trial leaks. For a set of trials $\mathcal{T}$ and a threshold of k words,

$$\text{ICR@}k = \frac{\left|\{\, i \in \mathcal{T}_k \;:\; \ell_i \geq k \,\}\right|}{|\mathcal{T}_k|}, \qquad \mathcal{T}_k = \{\, i \in \mathcal{T} \;:\; a_i \geq k \,\}, \tag{3.40}$$

where $\mathcal{T}$ is the set of trials being scored, i indexes a trial in it, ℓ_i is the number of leaked words on trial i, a_i is the number of words available to leak on trial i, k is the number of leaked words at which a trial counts as leaking, and $\mathcal{T}_k$ is the subset of trials with at least k words available to leak. Only trials with at least k available words are eligible, since a trial with a single available word can never reach ICR@2 and would only dilute the rate. This work uses $k = 2$: one shared content word can be coincidence, while two is a signal. The choice does not change any conclusion, as every k from 1 to 5 fully separates the no-processing and target-alone reference points (Section 3.4.7). The second rate asks how much leaks, as the share of the available words that reached the transcript, expressed as a percentage:

$$\text{mean leak} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \frac{\ell_i}{a_i}, \tag{3.41}$$

Lower percentages are better for both metrics.

3.4.4. Fabrication Rate

The Fabrication Rate (FR) highlights the case where the model has hallucinated a word that is neither present in the target or the interferer and is only computable with exact transcriptions

as given in LibriSpeech [21]. The FR metric is defined as follows: Over the set $\mathcal{T}$ of trials with a non-empty response, the statistic is the mean invented count per trial of the number of words in the model output that are not present in either the target or interferer,

$$\mathrm{FR}_{\mathrm{count}} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \left| \mathcal{I}_i \right|, \tag{3.42}$$

where $\mathcal{I}_i$ is the set of words in the i-th trial's output that are not present in either the target or interferer [AI7]. A rate of fabrication is also defined by whether a trial contains at least k invented words as defined in Equation 3.43.

$$\mathrm{FR@}k = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mathbb{1}\left[\left| \mathcal{I}_i \right| \geq k \right], \qquad k = 2 \text{ chosen in this work.} \tag{3.43}$$

where $\mathbb{1}$ is the indicator function. An invented word is not necessarily a hallucination as defined by [18]: a misheard word is counted as invented, but it is not a hallucination.

3.4.5. Deep Noise Suppression Mean Opinion Score

DNSMOS [12, 40] is a neural network that predicts what a collection of humans would have rated an audio clip on a scale of 1–5, where 1 is bad and 5 is excellent. Two variants are report: ITU-T P.835 [41], which gives speech quality (SIG), background quality (BAK) and overall quality (OVRL) and ITU-T P.808 [42], a single score that the REAL-TSE organisers adopted after entrants were found to game the P.835 metric.

3.4.6. Signal to Distortion/Interference/Artefact Ratios

SDR is used to measure how wrong the output audio is in total, and the SIR and SAR are used to measure in which direction the output audio is wrong. The SIR measures how much real audio that is not target speech is present in the output audio such as an interfering speaker or background noise, and the SAR measures how much of the output audio is not real audio, such as artefacts from the model. The output is split by orthogonal projection [43] into the target part s_{proj} of Equation 3.6, the interference e_{int} explained by the other speaker d and the noise, and the artefact e_{art} explained by none of the sources, so that $\hat{s} = s_{\mathrm{proj}} + e_{\mathrm{int}} + e_{\mathrm{art}}$:

$$\mathrm{SDR} = 10 \log_{10} \frac{\left\| s_{\mathrm{proj}} \right\|^2}{\left\| e_{\mathrm{int}} + e_{\mathrm{art}} \right\|^2} \tag{3.44}$$

$$\mathrm{SIR} = 10 \log_{10} \frac{\left\| s_{\mathrm{proj}} \right\|^2}{\left\| e_{\mathrm{int}} \right\|^2} \tag{3.45}$$

$$\mathrm{SAR} = 10 \log_{10} \frac{\left\| s_{\mathrm{proj}} + e_{\mathrm{int}} \right\|^2}{\left\| e_{\mathrm{art}} \right\|^2} \tag{3.46}$$

which are all measured in decibels (dB) where the higher ratio means better extraction of the target from the other audio in question.

3.4.7. Anchors

The following terms are used to describe the lower and upper bounds of the results section. The floor references passing the full mixture audio, unprocessed to a listener as the worst case scenario, and the ceiling references passing a clean target audio to the listener as best case. The ceiling system is perfect separation and the listener hears only the target audio. The floor and ceiling systems are not real systems, but rather reference points for the listener to compare against.

3.4.8. Uncertainty

Each 95 % interval is a percentile bootstrap [44]. The interval is computed using bootstrap resampling where the trials are redrawn with replacement 10,000 times, and on every draw the score is recomputed as the total errors over the total reference words [45]. Because trials from the same speaker are not independent, unique speakers are chosen rather than individual trials, and whole meetings for the AMI recordings.

4. Experiments

This section discusses the overall setup to conduct the experiments, including the evaluation protocol, and the experimental setup. The results of the experiments are presented in Section 5.

4.1. Training setup

The training setup is described below and considers how the training data is augmented, the optimisation and hardware used, how the final model is selected, the systems compared, and the actual training observed.

4.1.1. Data augmentation

The training data is augmented with varying strategies to prevent overfitting. The first strategy is to provide every trial twice per epoch where the target and interferer roles are swapped. This means the same mixture requires two different outputs where the only varying information is the enrollment clip. If a clip is only used once for a single speaker, the answer remains the same which can be memorised. Therefore the swap ensures that if the model memorises on one voice, it will be penalised if it fails to generalise to the other voice. The second strategy is to rotate enrollment clips for a target speaker. Each speaker per trial rotates between 3 enrollment clips so that the model does not memorise a single enrollment to a speaker. The third strategy is to randomly perturb the noise and reverberation for each trial per epoch. Finally, in the spirit of stochastic methods, a single random $4\,\mathrm{s}$ crop is taken from a random position in the trial for each epoch.

4.1.2. Optimisation and hardware

During training the AdamW optimiser [46] was used with a base learning rate of 0.0005, weight decay of 0.0001 and gradient clipping at 5.0. A batch size of 3 was used, thus 6 samples were

processed per iteration (see Section 4.1.1) with mixed precision training. The learning rate schedule included halving the learning rate when the Whisper WER stops improving. The target-absent term in the loss, L_{abs}, is off for the first 2 epochs, as the model in the early epochs would just output silence as this term is easy to optimise. The model is trained on Kaggle's Tesla T4 GPU for 12 hours per session and the random seed is set to 42 for reproducibility [AI5].

4.1.3. Model selection

An early stopping mechanism after 10 epochs without improvement was available as a backstop but never triggered. The baseline ran for one session of 14 epochs. The extension ran for 14 epochs, and because its in-training word error rate was still falling at the end of that session (its best epoch was the last), training was resumed for a second session, to 28 epochs.

The final model to base the evaluation on cannot simply be chosen by the model with the lowest validation loss as a result of the proxy training objective. The loss (Section 3.2.9) is a proxy for the downstream task, where the approach in the proxy is to reward better separation. However, better separation does not guarantee better recognition (Section 2.3): the epoch with the lowest loss for the extension has the highest LCF-WER of the chosen checkpoints as seen in Figure 4.1. Secondly, the model cannot be optimised directly against the LCF-WER as it is not differentiable. Finally a consideration is that the live judge is also too slow to run after every epoch. The model is thus chosen as follows: Whisper (paragraph 4.2.1) during training will create a shortlist of best candidate checkpoints based on the WER of the validation set. The candidates are then scored by the live judge (paragraph 4.2.1) on the validation set, where the lowest WER is chosen as the final model with ties broken by fewer invented words.

4.1.4. Systems compared

This work evaluates 3 models against the floor and ceiling (Section 3.4.7): baseline BSRNN (implemented following REAL-TSE [3]; Section 3.2), the extension BSRNN (Section 3.3), and WeSep [30]. There are four main reasons to include WeSep for comparison: *(i)* the REAL-TSE challenge [3], which this work uses as a guideline, builds its baseline system with WeSep; *(ii)* the WeSep model has the same BSRNN backbone with TF-Map cue of [27] as this work's baseline and extension, where the main architectural difference is WeSep's use of contextual embeddings rather than the single embedding per enrollment clip used by the extension, as discussed in Section 3.3.2; *(iii)* WeSep is used to validate the dataset constructed in this work, as WeSep was trained on clean Libri2Mix [9] and thus has not heard any of the speakers in the evaluation sets, so its extraction demonstrates what an established model can recover from these mixtures; and *(iv)* WeSep provides a stronger offline reference for extracting the target. The WeSep model itself is non-causal and almost four times the size of the baseline (27.2 M versus 7.19 M parameters), and thus shows what extraction can achieve when latency and model size are not a concern. WeSep was not trained on data that contained noise or reverberation, whereas trials in this work contain both. Secondly, WeSep itself has a causal separator however, the features are normalised over the entire clip before the separator. This setup means that every output

frame depends on audio that has not arrived yet and thus the model cannot be streamed and is treated as an offline system. Finally, the results of WeSep are not comparable to the results found in the REAL-TSE challenge [3] due to varying data, metrics and protocols.

4.2. Evaluation setup

The evaluation setup is described below and considers the listeners used, the latency measures, and the evaluation metrics. The evaluation metrics are described in Section 3.4 and the results of the evaluation are presented in Section 5.

4.2.1. Listeners

Offline ASR model. The offline transcriber is the `faster-whisper` implementation [47] of the Whisper [38] model, run with the `large-v3-turbo` checkpoint. This is Whisper large-v3 with its decoder pruned from 32 layers to 4 and then fine-tuned, giving 809 M parameters against the 1,550 M of large-v3 [48]. The large Whisper variants are a common choice for scoring how intelligible generated and extracted speech is [49, 50]. Decoding was greedy, with a single sampling temperature of zero and no conditioning on previously decoded text, so the same audio always produces the same transcript, and the language was fixed to English. The smaller `small.en` checkpoint (244 M parameters) is used only inside the training loop, where it transcribes a validation sample every epoch to shortlist checkpoints and schedule the learning rate.

Live judge. The live listener is `gemini-3.7-flash` [2], which was connected to via the `google-genai` Python package through the Application Programming Interface (API). The judge, like the ASR model, does not separate anything. The Gemini model is prompted to transcribe what it can hear in the extractor's output. The live model is the tool for which this work is being built and so its result is the most important with respect to the overall evaluation of the extractor. The prompt used can be found in Section E.1 and is deliberately worded to avoid instructing the judge to perform extraction. The judge is also prompted to provide a structured response with a status field alongside the transcript. The status field is used to indicate whether the judge heard speech or not, and if it did not hear speech, the transcript is left empty. This allows for a clear distinction for whether the model hallucinates as the status field can be used to determine whether the model heard speech.

Speech gating. The Silero VAD v6.2.1 [51] speech gate is implemented to filter out any audio that appears to contain no speech and acts as a pre-filter for the transcription tools. A clip without detectable speech is scored as an empty transcript without the need of calling the listener. This method is supported by [52] where the authors show that this method prevents inventing words on silence and mirrors what a deployed system would do. The gated output also shows missed words that a target said but was suppressed, and secondly shows correct output of silence when the target is absent.

4.2.2. Training curves

Both models were trained with the same loss (except the additional presence gate in the extension) and the same settings, but differ in how long they were trained. The baseline was trained for 14 epochs and the extension for 28. Figure 4.1 shows how the validation scores of each model changed during training. The top row of Figure 4.1 shows the breakdown of the loss for the target-present branch. The three terms of the loss, shown separately, highlight how the terms interact with each other. A typical trend seen for both baseline and the extension is that as the separation term L_{pres} for training decreases (rather sharply), the validation separation very gradually decreases, and then very gradually increases again. A positive note to be seen is how the multi-resolution spectrum term, L_{MR}, consistently decreases for both training and validation, showing how the extractor is able to reconstruct the audio with fewer artefacts. The gain term for ensuring that the model produces audio at a similar level to the target, L_{gain}, follows the same trend as the separation term, which is slight decrease followed by a slight increase. Finally, the baseline model is only trained for one session, as by the validation live judge results it can be seen that on the final epoch the validation WER has reverted back to a same score as the fourth epoch. Epoch 11 is chosen for the baseline as it has the lowest validation WER. Although separation is not the hallmark indicator of a good extractor for the context of the downstream task, when the term is decreasing in conjunction with the spectrum term, it is a good indication to continue training. The extension is thus trained for a second session as by the end of the first session, *(i)* the Whisper WER is generally decreasing although sporadic, and *(ii)* the separation term with the spectrum term is decreasing. Epoch 21 has a similar validation WER to epoch 27 (39.59 % against 39.23 %), and is preferred when considering the breakdown of the validation scores which is found in Appendix F. Epoch 27 in 95–97 % of resampled validation sets causes Gemini to invent more words per trial (2.56 against 2.20) and to invent two or more words more often (55.5 % against 46.9 %) than epoch 21. The tradeoff is that epoch 27 almost always leaks less interfering speech in a trial (20.2 % against 23.1 %). Since the difference between the leftover speech and introduced artefacts is equally as strong (since the WER is similar) in this validation set, the choice is made to prioritise less invention as this will translate to more unseen trials. Leakage is target and interferer dependent, but artefacts will be produced whenever the model is used. Finally, when considering the presence gate, it can be seen that the loss for the gate prior decreases steadily for both training and validation with an increase after epoch 25. This shows that the model decision for the gate to turn off or on matches the prior's speech and silence durations. However, a concerning note is in the loss for the gate labels. The label loss decreases from from 15 to 4 in training while validation stays flat between 12.5 and 18.5, so the gate memorised when the training speakers talked. The gate is what allows speech to be passed through to the downstream model, and with the oscillation in the validation loss, it is clear that the model had not learnt anything in the training set that generalises to the validation set. Therefore, the gate is not used in the final model and the extension is evaluated without it.

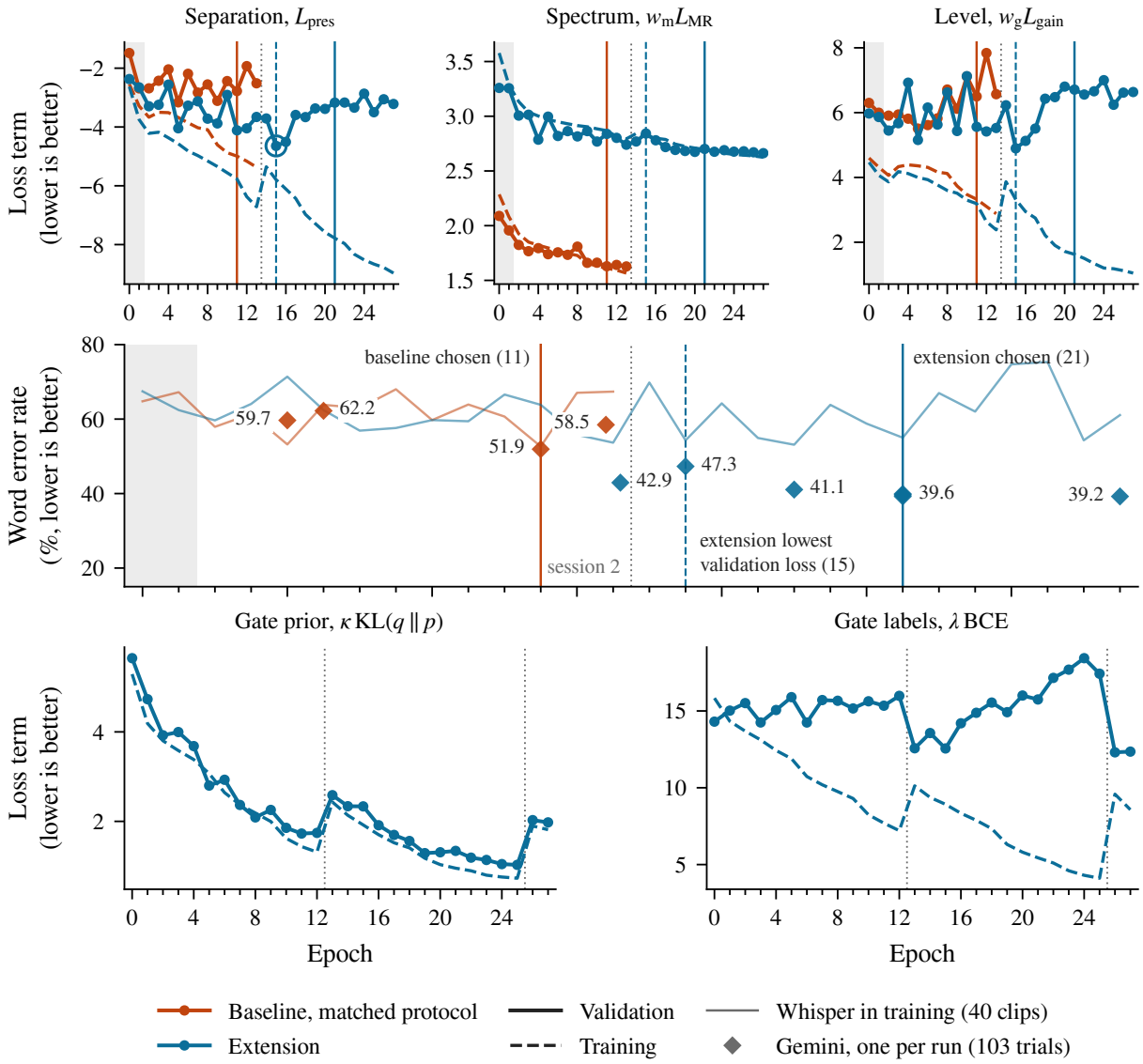


Figure 4.1: The baseline (orange) and the extension (blue), trained under the same protocol. The top row shows the three weighted terms of the target-present loss which are attributed to target specific components, as seeb in the first bracket of Equation 3.11. The middle row shows the word error rate on the 103 validation trials with both speakers present. Each diamond is one run of the live judge on the same audio. The solid vertical line marks the chosen epochs, and the grey dotted line where training resumed in a second session. The bottom row shows the loss curve of the presence gate, which is only trained on the extension.

4.2.3. Latency measures

Latency is measured against a 200–300ms end-to-end buddet. There are two components to consider: time to consume and time to process the audio. The real-time factor (RTF) is the ratio of the time taken to process the audio to the length of the audio, and for a model to be able to keep up with real-time audio, the RTF must be less than 1. The second component considers the latency created by the architecture itself which cannot be improved by faster hardware. To produce the output frame for a given window, the model must first have received that entire window. This forces a delay of the processing of the window as per Section 3.2.2.

5. Results

5.1. Why the metric is needed

The distinction between Section 3.4.3 and Section 3.4.4 matters when investigating the insertion and substitution error of the LCF-WER metric. With Gemini as the listener, the fabrication and leakage error trade against each other. Two fine tunes were run on the extension where 6 additional epochs were trained. As shown in Table 5.1, the interferer weighted fine tune shows the clear reduction in how much the interferer speaker's content is leaked into the output, but at the expense of more invented words (as more artefacts are introduced hence more hallucination). This is shown by the mean percentage of the interferer words leaked into the output improving by 3.26 percentage points, but the number of invented words increasing. The artefact weighted fine tune shows the opposite situation with a heavy increase in leaked words but slight reduction in invented words. Each error has a different cause and solution, and thus should be treated separately, which would be hidden by the LCF-WER metric alone.

Table 5.1: Effect of weighting one source of error during fine-tuning, with Gemini as the listener. Lower is better in every row. Negative values are improvements.

	Leak weight		Artefact weight	
	change	95 % CI	change	95 % CI
LCF-WER ↓	$+0.47$	$[-3.20, +4.15]$	$+9.34$	$[+3.65, +15.37]$
mean leak ↓	-3.26	$[-6.59, -0.03]$	$+14.66$	$[+9.09, +20.52]$
invented words per trial ↓	$+0.72$	$[+0.27, +1.21]$	-0.20	$[-0.65, +0.25]$

5.2. Content fidelity

The content fidelity metrics are used to measure how the live artificial intelligence system is able to interpret the audio after the extraction process. The live system is a black box, non-deterministic system, the stability of the interpretation of the audio is also important to consider. The sections to follow consider the content fidelity for a live judge on both real and the simulated data.

5.2.1. The live judge

The live judge results are shown in Table 5.2. These results are most indicative of the overall performance of the extractor system as the live judge is the end user of the extraction process. To better understand the variability of the live judge on the same audio, Figure 5.1 shows how the different systems changed in the LCF-WER metric across three independent runs.

The baseline improves the LCF-WER by 9.68 points, which is a 15.4 % relative improvement over the floor. There is a decisive improvement however in the baseline model's performance over the floor with regard to including an interfering speaker's content, as the average percentage of interfering words included in the output is 17.15 points lower than the floor. This came at a trade-off with the number of hallucinated words. The baseline system subsequently has

Table 5.2: Content recovered by Gemini. `gemini-3.7-flash`, audio in and text out. Each value is the average of 3 runs on the same 103 trials, and ± is the standard deviation of those three run scores. 95 % CI is the range for the 3-run LCF-WER in percentage points, redrawing whole speakers (Section 3.4.8) and Figure 5.1 shows how much the runs differ. Lower is better in every column.

system	LCF-WER ↓ (%)	95 % CI	ICR@2 ↓ (%)	mean leak ↓ (%)	invented ↓ /trial	FR@2 ↓ (%)
Floor	62.94 ± 0.37	[56.02, 70.18]	75.73 ± 0.00	63.90 ± 0.65	1.27 ± 0.03	30.7 ± 3.1
Baseline	53.26 ± 1.82	[46.23, 60.35]	58.58 ± 2.02	42.20 ± 0.87	2.52 ± 0.20	49.8 ± 1.5
Extension	39.59 ± 0.34	[33.35, 45.92]	33.66 ± 2.02	23.08 ± 0.97	2.20 ± 0.11	46.9 ± 2.0
WeSep	**25.80** ± 0.56	[20.71, 31.14]	**25.24** ± 0.97	**13.04** ± 0.32	**1.80** ± 0.02	**38.6** ± 2.3
Ceiling	1.19 ± 0.18	[0.80, 1.60]	0.00 ± 0.00	0.00 ± 0.00	0.22 ± 0.03	0.3 ± 0.6

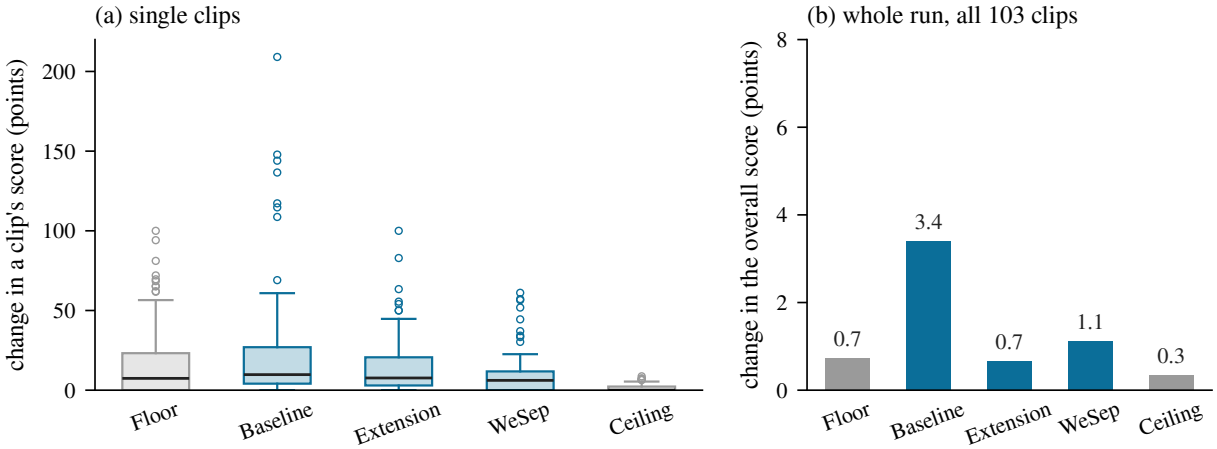


Figure 5.1: How much Gemini's LCF-WER changed across the three runs in Table 5.2, where change is the highest of the three run scores minus the lowest. (a) The box plot showing how the change varies per clip over the averaged runs. A clip's score can pass 100 % and can occur when Gemini adds words through hallucination or by including interfering speech. (b) Change in the overall score over all 103 clips. Floor and ceiling (grey) are reference points.

fabricated/invented more words on average per trial as seen in the table. Finally, the baseline system also more frequently fabricates, as indicated by FR@2. Therefore, both the frequency and quantity of fabrication have increased over passing the audio unprocessed.

The baseline system is a clear improvement over the floor, which is indicative that the extractor has begun to find a signal in the mixture that is associated with the target speaker. For the extension, the inclusion of the pretrained ECAPA speaker encoder and passing the enrollment audio in parts instead of a single embedding has markedly improved the performance of the extractor. The LCF-WER has improved by 13.67 points over the baseline, which is a 25.7 % relative improvement. The extension has both reduced the share of leaked words per trial and the frequency of leakage by substantial amounts. Unlike the leakage, the extension did not drastically invent less words per trial, with only a slight improvement from 2.52 to 2.20 words per trial. However, both baseline and the extension invent well above the floor which is at 1.27 words invented per trial.

Figure 5.1 demonstrates the stability of the LCF-WER across the three independent runs

measured per clip. The baseline demonstrates the worst stability, changing by 1.82 points, with greater variability as compared to floor. The extension has the best stability even when compared to WeSep. On single clips, there are outlier performance changes of up to 209 points (when one run results in 209 more errors per 100 reference words than another run), these outliers could be incredibly difficult cases that are not indicative of the overall performance of the system. However, as seen in the bar plot, the extension on average has the smallest changes in overall performance across the three runs, although performs much worse on outlier cases as seen in the box plot.

Table 5.3: Gemini's response in each trial case of Section 3.1.3. Cases 1 and 2, where the target spoke, report LCF-WER. Case 3, where only the other speaker spoke, reports the words Gemini output per trial, where 0 is correct, and how many outputs were silenced. Case 4 (noise only, $n = 8$) is omitted where no system produced a word, and every output was below the speech gate.

	Case 1: both	Case 2: target only	Case 3: other speaker only	
system	LCF-WER ↓ (%, $n = 103$)	LCF-WER ↓ (%, $n = 47$)	words/trial ↓ ($n = 42$)	silenced ↑ (of 42)
Floor	62.94	1.65	30.69	0
Baseline	53.26	2.48	24.12	8
Extension	39.59	**3.56**	**16.50**	**18**
WeSep	**25.80**	5.40	17.95	7
Ceiling	1.19	1.46	—	—

Table 5.3 highlights where the extension shines. For the cases where the extractor needs to either suppress only background noise surrounding a target or only interfering speech, the extension performs better than WeSep. Secondly, if the extractor needs to suppress background noise such that a voice activity detector (VAD) does not detect any speech, the extension is able to do this more often than WeSep getting 18 noisy clips suppressed as compared against WeSep's 7. The improvement in the extension's performance can be attributed to the diversity of training examples that include those specific cases. Interestingly, all the models perform worse on the target-only case than just leaving the target unprocessed. This is likely due to the robustness of Gemini to background noise, leaving the extractor model to introduce artefacts that are misinterpreted by Gemini. Overall, the extension has shown improvement over WeSep in different acoustic scenarios, although it still falls short when considering pure speech extraction with multiple speakers.

5.2.2. Real recordings

Table 5.4 shows Gemini's results on the AMI meeting recordings of Appendix C. The conclusion is the same as for the simulated data: the extension improves the LCF-WER over doing nothing, although the offline WeSep system performs better for the strict extraction task. WeSep has the lowest LCF-WER (80.56 %), followed by the extension (91.26 %) and then no processing (115.51 %). A demonstration of how the model distorts the speech can be seen by the target

words missed metric, where Gemini is able by itself without any processing to only miss 3.46 % of the target words, but even with WeSep, Gemini misses 27.71 % of the target words. This indicates how the extractor degrades the target speech at the cost of removing the background interfering speech. This is due to the artefacts introduced by the extractor, as seen in the increase in invented words per trial (1.48 to 1.83) and rate of fabrication (38.33 % to 49.46 %) for the extension.

Table 5.4: Content recovered by Gemini on 300 real two-speaker stretches of AMI meeting recordings (Appendix C). Lower is better in every column.

	No processing	Extension	WeSep	Target's headset
LCF-WER (%) ↓	115.51	91.26	**80.56**	72.53
95 % CI	[103.95, 128.89]	[84.29, 98.87]	[73.49, 88.32]	[62.52, 83.72]
target words missed (%) ↓	3.46	45.25	**27.71**	2.31
ICR@2 (%) ↓	89.14	33.71	**27.72**	58.80
mean leak (%) ↓	82.80	25.19	**19.64**	51.96
invented /trial ↓	1.48	**1.83**	2.65	1.13
FR@2 (%) ↓	38.33	**49.46**	64.17	30.10
silenced (of 300) ↓	0	110	**38**	0

5.3. Conventional metrics

The three kinds of separation considered in this project are the target from the background noise (distortion), interference from other speakers and finally from artefacts introduced by the extractor system. The signal-domain separation results are shown in Table 5.5. These metrics highlight what the extractor system is doing in the signal domain and can best be used to understand the trade-offs between the different types of separation. A note about the values in the table: SAR for the floor and ceiling are both at +30 dB, since these systems do not contain a model there can be no artefacts introduced. The extension has significantly

Table 5.5: Signal-domain decomposition, in dB, and DNSMOS predicted opinion scores (personalised model, from 1 to 5) on the same 103 trials. Higher is better in every column. The $30\,\mathrm{dB}$ entries are the cap created by the denominator floor $\tau = 10^{-3}$ (Section 3.4.6), not measurements.

	Signal (dB)			DNSMOS (1–5)			
system	SDR ↑	SIR ↑	SAR ↑	P.808 ↑	SIG ↑	BAK ↑	OVRL ↑
Floor	-1.12	-1.12	$+30.00$	2.913	4.090	2.031	2.497
Baseline	$+1.44$	$+5.43$	$+8.58$	2.995	2.961	2.579	2.144
Extension	$+2.29$	$+9.17$	$+6.28$	**3.030**	2.957	**3.045**	2.306
WeSep	$+4.68$	$+10.34$	$+8.26$	3.010	**3.450**	2.640	**2.510**
Ceiling	$+30.00$	$+30.00$	$+30.00$	3.550	4.175	3.592	3.429

improved the SIR metric over the baseline, and is comparable to the WeSep model. The gap in performance between the extension and WeSep therefore seems to lie in the SAR and SDR metrics. Therefore when considering how to improve the extension, the next iteration should

potentially target mechanisms that reduce the artefacts introduced by the extractor system as artefacts are hypothesised to increase the fabrication rate. The baseline has the highest model SAR metric. This should not be attributed to the model being better at reducing artefacts, but rather due to the simplistic nature of the model design. The baseline model is not as aggressive in masking out the interfering speaker's content, and therefore does not introduce as many artefacts into the output.

Considering P.808 in isolation, the extension has a comparable overall quality score to WeSep. By the one perceptual quality metric the extension produces audio that is as pleasing to a human listener (by the P.808 model standards) as the WeSep model. When considering the other three metrics, the WeSep model has a better SIG and OVRL score than the extension. The extension scores lower than the floor for both the SIG and OVRL metrics, and yet as seen in the content fidelity results, the extension has improved the content fidelity over the floor by a significant margin. This demonstrates that the perceptual quality metrics are not good indicators for the purposes of this research project. The metric in itself is subjective to the actual perceptual system and does not provide indication that what sounds good to a human listener is necessarily better for the content fidelity of the target speaker's audio.

5.4. Latency and throughput

Table 5.6: Baseline and extension figures are the median of three runs each, alternated in one session on an otherwise idle machine, with the range in brackets. The requirement is RTF < 1 *and* latency within 200–$300\,\mathrm{ms}$. The first is the binding constraint, because a chunk that finishes inside its own duration meets the latency budget automatically.

	RTF ↓		latency (ms) ↓		meets
system	mean	p99	mean	p99	budget
Baseline	0.69 (0.66–0.70)	0.79 (0.79–0.86)	175 (173–176)	183 (183–189)	yes
Extension	0.69 (0.68–0.71)	0.78 (0.77–0.82)	175 (175–177)	182 (182–185)	yes
WeSep	2.854	5.300	348.3	544.0	no

The latency and throughput results are shown in Table 5.6. The baseline and the extension are able to meet the real-time factor and latency requirements for a live system. The WeSep model is not able to meet the requirements and therefore cannot be used in a live system. Furthermore, the extension did not miss any chunks and the worst case delay was 196 milliseconds, which is below the 200–300 millisecond requirement.

6. Conclusion

The work in this report set out to build a target speaker extractor that could prepare audio for a live speech-to-speech model under the constraints of being able to run in real time, causally and on a single channel of audio. The extractor was judged by what Gemini recovers from the extracted output, and not by the actual separation quality of target speech from the other audio.

All of the findings relate to at most two-speaker mixtures, and optimised and tested for Gemini.

Key findings. The simple word error rate hides the trade-off that matters for the downstream model. The word error rate alone does not discern how insertions or substitutions are obtained, and both are rooted in different causes. The causes include either invention or leakage, and an extractor needs to tradeoff optimising one against the other. The next finding includes the confirmation for how signal, perceptual and offline measures do not correlate with the live model's perception of the audio. Despite being needed to train an extractor by forming loss terms from the metrics, the metrics can only act as a proxy for how the model extracts and not necessarily how well the extractor performs for the downstream model. The next finding is that telling the extractor more about the target's voice is what helped the most. This involved considering the enrollment and the conditioning of the extractor on the target speaker's embedding. The cue was split into parts and adding the speaker embedding lowered Gemini's word error rate from 53.3 % for the baseline to 39.6 %. The final finding is that the live model does not have much tolerance for artefacts where even with the extension Gemini invented 2.20 words per trial where doing nothing only 1.27.

Future work. The structure and nature of the approach formulated in this work allows for a number of future directions to be explored. The first suggestion is considering the streaming architecture itself. The LSTM layers of the extractor could be replaced with state space layers. A state space layer is a recurrent layer whose updates are linear, which allows for parallel training while still running frame by frame. SA-Mamba [53] applies this idea within a band-split, dual-path design similar to the one in this work, and conditions the state space layers on the target speaker. It placed second in the online track of the REAL-TSE challenge, while its report describes training on simulated mixtures only, and the organisers attribute most of the gains of the top systems to their data [3]. The architecture has not yet been used with a live speech model as the listener and secondly it did not have a training recipe built around one, as in this work. The next suggestion would be to consider the current objective function. It is a proxy for general extraction rather than for the downstream task. Weighting the leaked speech or the artefacts in the loss moved errors from one source to the other but did not lower the LCF-WER (Section 5.1), so an objective closer to *how* Gemini listens is still needed. Another idea would be to address how the extractor struggles with the trade-off between removing interfering speech and introducing artefacts. A likely reason is the aggressive masking, which can leave sharp transitions where the interfering speech was removed. The extractor could instead be trained to also extract the interfering speaker, so that it learns what to remove as well as what to keep. Rather than leaving gaps where the interfering speech was removed, the extractor could fill them with a signal that Gemini ignores, such as low-level noise, leaving nothing for Gemini to fill with words. Gemini, during experimentation, was found to be robust to very low-level noise, but the idea is to make audio clips that are optimised to be ignored by the listener. Finally, I believe there is potential in the presence gate, which was only trained on the extension. I would

suggest that the gate rather be trained on the two speaker condition rather than a general target present condition. Right now the gate only theoretically helps for interferer only trials, but this should be modified to help prevent leakage in the two-speaker condition. This could be in the form of allowing the gate to close on the interferer's frames when the target is present.

Plagiarism and AI declaration

During the course of this project, I made use of Claude Code to assist me in the research process. It was used in general cases where redundant tasks / code was needed (soft use) and then cases where a citation for the tool is required (hard use) in accordance to the guidelines set out in "Guidelines for using GenAI tools in Learning and Assessment". Listed below are all the specific items that I used the tool for:

AI1. **Research:** I prompted Claude Code to assist me in finding relevant research papers and articles that may have potentially solved the TSE task. I used the tool to highlight short comings and advantages of different approaches. It was from this research that I decided on the BSRNN model. Specifically for the extension of the BSRNN model, I prompted the tool to analyse the short comings of the model and to suggest potential avenues to explore to improve the model. I then decided on the specific design choices for the extension from my own understanding of the tool's response.

AI2. **Writing assistant:** I wrote the report myself and made use of Claude Code in fixing grammar, spelling and sentence structure. The tool was not given access to writing directly to the report, but rather would suggest corrections if a claim or sentence was not clear or inaccurate. I would ask the tool if my summary of a paper or method was fair and accurate to not misrepresent the work of others or my own work. The formulations and equations for the described methods were sourced by the original authors, but I would first rewrite them from my understanding and then ask the tool to help change the notation to be consistent with the rest of my report.

AI3. **Diagrams:** Figure 3.2 specifically I prompted Claude Code to assist with the TikZ code for the best layout of the diagram. The other figures in the report were reformatted and editted by Claude Code to have a scientific style and to be consistent with the rest of the report.

AI4. **Code (soft):** I made use of Claude Code to assist me in documenting the decisions from my discussion with the tool into the code and recorded them seperately. All decisions were logged into the documentation and can be found in the "'docs/decisions"' directory in the repository found at https://github.com/grantxxcoder/TargetSpeakerExtractor. For the code pertinent to the baseline and extension model for the actual TSE task, I made use of Claude Code to debug and fix errors.

Furthermore, for the baseline, because it needed to follow the architecture found in the REAL TSE paper, I needed assistance in implementing the architecture as described in the paper. I prompted the tool to verify that the implementation was consistent with the paper and to suggest changes if it was not.

AI5. **Code (hard):** I orginally wrote my code in scripts but Kaggle only accepts the files in a notebook format. Thus I prompted the tool to fully convert my scripts into a bundle of configuration files and notebooks that can be run on Kaggle. The tool then created multiple hyperparameter configuration files (*.yaml*), notebooks for training different models and dataset loaders. I then prompted the tool to verify that the notebooks were consistent with the scripts and that they would run on Kaggle. The generated files only contain the code required to train and save the models and no additional logic is used to change the model or training procedure.

AI6. **Equations and Formulations:**

(a) For the first extension, the splitting the cue into parts idea came after the initial diagnosis of the failures of the BSRNN. I prompted Claude Code with ideas for improving the speaker representation in which the splitting of the cue into parts was suggested. I used Claude to verify the equations and to ensure consistent notation.

(b) I had the idea of using a two state target present markov model as a conditioning gate for the BSRNN model. I was able to derive the expressions by hand until the KL term myself. For the KL term, Claude Code pointed out that under the mean field assumption the expectation over the neighbouring frames splits into the product of their probabilities and it cited this identity to the paper by Ghahramani and Jordan [34]. I then worked through each step myself. For the training and inference, Claude Code suggested and derived the equations for the Monte Carlo sampling and the binary Concrete distribution.

AI7. **Created Metrics:** The new metrics, the fabrication rate and the interferer rate, were my own ideas, in which I only used Claude to help formulate the equations.

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A. Room construction

The acoustics are a controllable variable rather than a property of whichever room happened to be recorded. Rooms are rectangular (“shoebox”) and the room accoustics are simulated using `pyroomacoustics` 0.10.1 [23]. For every trial the renderer draws the room’s three dimensions, a reverberation time, and independent positions for the microphone and the two talkers. Wall absorption is not drawn directly but solved using the Sabine formula and the draw is rejected and retaken if the combination is not physically possible. Both talkers are placed in the same room and share the microphone.

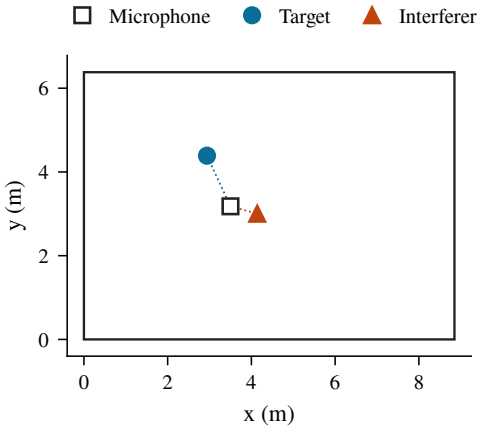


Figure A.1: Plan view of the room rendered for trial `train-42-007495`. The room measures $8.85 \times 6.38 \times 3.14\,\mathrm{m}$ ($177\,\mathrm{m}^3$) with a reverberation time of $0.250\,\mathrm{s}$, for which the Sabine relation gives a wall absorption of 0.5479. The microphone sits at $(3.50, 3.18, 1.28)$ m, the target at $(2.94, 4.39, 1.66)$ m and the interferer at $(4.14, 3.00, 1.57)$ m, placing them $1.39\,\mathrm{m}$ and $0.72\,\mathrm{m}$ from the microphone respectively.

B. Construction parameters

Every parameter below is drawn independently per trial from the stated distribution, except where a regime overrides it. The table gives the training and validation settings. The `eval_public` and `eval_private` splits draw SIR from $[-5, 15]$ dB and use no regimes. Ranges written $[a, b]$ are sampled uniformly on that interval.

Parameter	Distribution	Meaning
Trial layout		
Mixture length	$[15.0, 20.0]$ s	Duration of the rendered trial
Overlap ratio	$[0.0, 0.7]$	Fraction of target speech with the interferer also talking
Overlap tolerance	0.03	Accepted deviation from the requested overlap
Target activity ratio	$[0.15, 0.78]$	Fraction of the trial in which the target speaks

Continued on the next page

Parameter	Distribution	Meaning
Activity tolerance	0.1	Accepted deviation from the requested activity
Utterances per source	≤ 6	Cap on how many utterances one recording may contribute
Levels		
SIR	$[-10.0, 10.0]$ dB	Target loudness relative to the interferer
SNR	$[0.0, 20.0]$ dB	Target loudness relative to the noise bed
Target loudness	$[-33.0, -25.0]$ LUFS	Absolute level of the target
Clip ceiling	0.95	Peak after which the whole trial is rescaled together
Room		
T_{60}	$[0.25, 0.6]$ s	Reverberation time
Room length, width	$[5.0, 10.0]$ m	Floor plan
Room height	$[3.0, 4.0]$ m	
Source distance	$[0.1, 2.0]$ m	Talker to microphone
Source height	$[0.9, 1.8]$ m	Talker, seated to standing
Microphone height	$[0.9, 1.8]$ m	
Early reflections	50 ms	Cutoff separating early echoes from the late tail
Enrollment		
Enrollment length	5.0 s	Single clip, levelled, carrying no room
EQ augmentation	$p = 0.5$	Three peaking biquads, log-uniform $120\,\mathrm{Hz}$ to $6400\,\mathrm{Hz}$, $\pm 6\,\mathrm{dB}$, $Q = 1.0$, RMS restored
Speaker pairing		
Same-gender fraction	0.5	Proportion of trials whose two talkers share a gender
Speech detection [51]		
Silence threshold	250 ms	Pause length after which a talker counts as stopped
Speech probability	0.5	Frame-level threshold
Minimum speech	100 ms	Shortest run counted as speech
Speech padding	30 ms	Added either side of a detected run
Noise-bed rejection	0.5 s	Longest run of detected speech tolerated in a WHAM! clip
Regimes		
Regime weights	0.6 / 0.4	Proportion of trials drawn as `base` / `hard`

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Parameter	Distribution	Meaning
`base` SNR	$[8.0, 20.0]$ dB	Regime override
`base` activity	$[0.45, 0.78]$	Regime override
`base` T_{60}	$[0.25, 0.5]$ s	Regime override

Table B.1: Construction parameters and their sampling distributions.

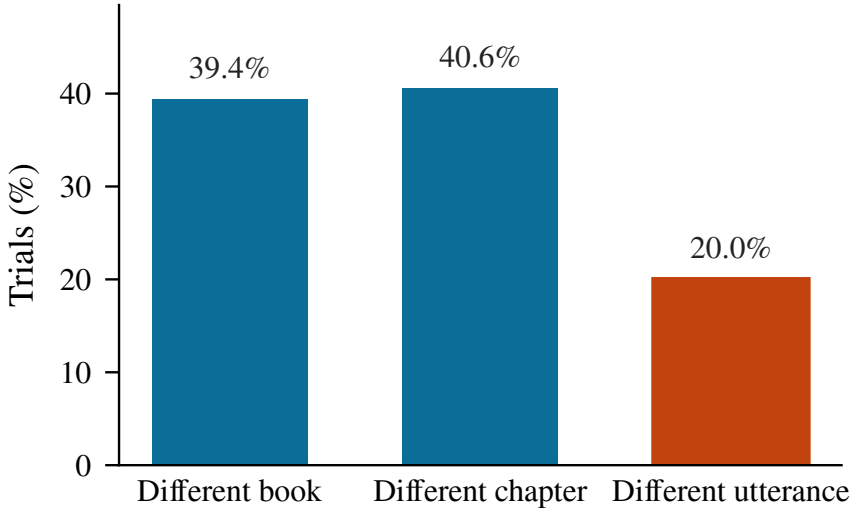


Figure B.1: Distribution of the source of the enrollment clips across the dataset.

C. AMI evaluation set

Property	Value
Recordings	
Source	AMI Meeting Corpus, scenario meetings [25]
Trials	300, totalling 77.9 min
Trial length	Median 15.8 s, range 10.1–20.0 s
Rooms	3, one per recording site (Edinburgh, IDIAP, TNO)
Meetings	58
Model input	One table microphone, level as recorded
Target alone	The target's headset and it also picks up the other speaker
Enrollment	5.0 s of the target speaking alone, from their headset in a different meeting; 247 distinct clips
Speakers	

Continued on the next page

Property	Value
Unique speakers	60
Trials per target	Median 5, range 1–12
Sex	45 male, 15 female
Pairing	236 same-sex trials, 64 mixed-sex
Native English	21 of 60 speakers; 108 of 300 trials have a native-English target
Age	Mean 26.8, median 25.5, range 19–58 years
Content	
Target words	Median 14 per trial, range 5–52
Other speaker's words	Median 15.5 per trial, range 5–70

Table C.1: The AMI evaluation set

D. Presence gate

D.1. Markov prior

The transition matrix of the Markov prior in (3.21), set by (3.24). Rows are the state in frame $t-1$ and columns the state in frame t, with 0 for the target silent and 1 for the target speaking:

$$A = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} = \begin{bmatrix} 1 - 1/\bar{d}_{\text{off}} & 1/\bar{d}_{\text{off}} \\ 1/\bar{d}_{\text{on}} & 1 - 1/\bar{d}_{\text{on}} \end{bmatrix} = \begin{bmatrix} 0.99796 & 0.00204 \\ 0.00168 & 0.99832 \end{bmatrix}, \tag{D.1}$$

with $\bar{d}_{\text{on}} = 596.34$ frames ($4.77\,\text{s}$) and $\bar{d}_{\text{off}} = 489.53$ frames ($3.92\,\text{s}$), measured over the 9955 training trials with each speaker in turn as the target, as in training.

E. Evaluation protocol details

E.1. Prompt for live-model transcription

```
Write down the words spoken in this audio, exactly as spoken.

If no words are audible, leave the transcript empty and set status to no_speech.
```

E.2. Pinned components of the measuring instrument

Component	Identity
Text normaliser	`whisper-normalizer` 0.1.15, `EnglishTextNormalizer` [37, 38]
Baseline transcriber	`faster-whisper` 1.2.1, `large-v3-turbo`, int8 quantisation, greedy decoding [38, 47, 48]
In-loop transcriber	`faster-whisper` 1.2.1, `small.en`, int8 quantisation, greedy decoding
Word error rate	`jiwer` 4.0.0 [54]
Stopword list	`NLTK` English stopwords [39]
Judge model	`gemini-3.7-flash`
Random seed	42

Table E.1: Components of the evaluation instrument.

F. Choice of the extension's epoch

Epochs 21 and 27 of the extension were both judged by Gemini on the 103 two-speaker validation trials (Figure 4.1) and tie on word error. Table F.1 compares them on every Gemini measure of Section 3.4.

Table F.1: Epoch 21 (chosen) against epoch 27 (the last) of the extension, on the 103 two-speaker validation trials. Lower is better in every row.

measure	Epoch 21	Epoch 27	Difference (21 − 27)	95 % CI	Epoch 21 better (% of draws)
LCF-WER (%)	39.59	39.23	$+0.36$	$[-3.00, +3.78]$	41.1
ICR@2 (%)	33.66	32.04	$+1.62$	$[-5.18, +8.41]$	31.5
mean leak (%)	23.08	20.16	$+2.93$	$[-0.47, +6.42]$	4.6
invented /trial	2.20	2.56	-0.37	$[-0.79, +0.08]$	94.9
FR@2 (%)	46.88	55.45	-8.56	$[-16.83, 0.00]$	97.4